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Muon decay in the Standard Model Effective Field Theory

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Abstract

The Standard Model Effective Field Theory (SMEFT) is an extension of the Standard Model of elementary interactions supplied with a set of higher dimensional gauge invariant operators, here considered up to mass dimension 6. Assuming that effects of so-called 'New Physics' can be parametrized by these operators, one gets the new vertices of interactions between the Standard Model fields. As a result, additional Feynman diagrams must be included while performing calculations within the SMEFT. An important elementary process which is affected by these operators is muon decay into electron and neutrinos. In the Standard Model there is only one possible diagram with W boson exchange contributing to this decay, while in SMEFT there are in total 4 possible such diagrams. In this thesis muon decay rate Γ_μ and Fermi coupling constant G have been obtained in terms of the SMEFT operators, including some contributions neglected previously in literature.

Key words

muon decay, Standard Model Effective Field Theory, Fermi coupling constant

Area of study (codes according to Erasmus Subject Area Codes List)

13.2 Physics

The title of the thesis in Polish

Rozpad mionu w efektywnym rozszerzeniu modelu standardowego

Contents

Chapter 1

Introduction and Motivation

The Standard Model (SM) is a well established theory confirmed with astonishing accuracy. However, there are indications of underlying fundamental theory that goes beyond Standard Model Electroweak scale $v \sim 246$ GeV. Without knowledge of what fundamental theory is, the Standard Model Effective Field Theory (SMEFT), defined as SM extended with complete set of gauge invariant operators constructed out of the SM fields, allows to parametrize effects of Physics beyond the Standard Model. Assuming that Λ is energy scale of the SM extension, all 'New Physics' observable phenomena are suppressed by v/Λ powers. Despite the fact that the SMEFT does not indicate clearly what fundamental theory is, it gives hints where new phenomena may be hidden. In order to look for 'New Physics', one may compare processes and quantities well known within the SM with those obtained within SMEFT. One such process is the muon decay, used to derive from the experiment the value of Fermi constant G. Calculating muon decay rate Γ_μ and its lifetime τ_μ allows to find corrections to G depending on Wilson coefficients of SMEFT operators. In this thesis we include new operators up to dimension 6 (i.e. up to $\mathscr{O}(1/\Lambda^2)$ order).

1.1. SMEFT Lagrangian

Notations and conventions are adapted from [1], therefore SM matter content takes form presented in Table 1.1:

Table 1.1: SM matter content in gauge basis

	fermions					scalars
field	$l'^{j}_{Lp} = \begin{pmatrix} \nu'_{Lp} \\ e'_{Lp} \end{pmatrix}$	e'_{Rp}	$q'^{\alpha j}_{Lp} = \begin{pmatrix} u'^{\alpha}_{Lp} \\ d'^{\alpha}_{Lp} \end{pmatrix}$	u'^{α}_{Rp}	d'^{α}_{Rp}	$\varphi^j = \begin{pmatrix} \varphi^+ \\ \varphi^0 \end{pmatrix}$
hypercharge Y	$-\frac{1}{2}$	-1	$\frac{1}{6}$	$\frac{2}{3}$	$-\frac{1}{3}$	$\frac{1}{2}$

where $j = 1, 2$ are isospin, $\alpha = 1, 2, 3$ colour and $p = 1, 2, 3$ generation indices. Full SMEFT Lagrangian up to $\mathscr{O}\left(\frac{1}{\Lambda^2}\right)$ corrections reads:

$$\mathscr{L}_{SMEFT} = \mathscr{L}_{SM} + \frac{1}{\Lambda} C^{\nu} Q^{(5)}_{\nu} + \frac{1}{\Lambda^2} \sum_{l} C^{l} Q^{(6)}_{l} + \frac{1}{\Lambda^2} \sum_{f} C'^{f} Q'^{(6)}_{f} + \mathscr{O}\left(\frac{1}{\Lambda^3}\right), \tag{1.1}$$

where $Q'^{(6)}_{f}$ refers to operators containing fermion fields, $Q^{(6)}_{l}$ to all other dimension 6 operators and $Q^{(5)}_{\nu}$ to the only dimension-5, lepton-number violating operator which reads [2]:

$$Q_{\nu\nu} = (\tilde{\varphi}^{\dagger} l'_{Lp})^T \mathbb{C} (\tilde{\varphi}^{\dagger} l'_{Lr}), \tag{1.2}$$

where $\mathbb{C} = i\gamma^2\gamma^0$ is charge conjugation matrix and $\tilde{\varphi} = iT^2\varphi^{\star}$. From this point, cut-off scale Λ is absorbed in the definitions of Wilson coefficients, i.e.: $C^{(5)}_{X}/\Lambda \to C^{(5)}_{X}$, $C^{(6)}_{X}/\Lambda^2 \to C^{(6)}_{X}$.

Standard Model part reads:

$$\begin{aligned}\mathscr{L}_{SM} = &-\frac{1}{4}G^A_{\mu\nu}G^{A\mu\nu} - \frac{1}{4}W^I_{\mu\nu}W^{I\mu\nu} - \frac{1}{4}B_{\mu\nu}B^{\mu\nu} + (D_\mu\varphi)^\dagger(D^\mu\varphi) + m^2\varphi^\dagger\varphi \\ &-\frac{1}{2}\lambda(\varphi^\dagger\varphi)^2 + i(\overline{l'_L}\not{D}l'_L + \overline{e'_R}\not{D}e'_R + \overline{q'_L}\not{D}q'_L + \overline{u'_R}\not{D}u'_R + \overline{d'_R}\not{D}d'_R) \\ &-(\overline{l'_L}\Gamma_e e'_R\varphi + \overline{q'_L}\Gamma_u u'_R\tilde{\varphi} + \overline{q'_L}\Gamma_d d'_R\varphi + h.c.),\end{aligned} \tag{1.3}$$

where primed symbols refer to fields and coefficients in gauge basis, while unprimed will denote those in mass basis.
Covariant derivative reads:

$$D_\mu = \partial_\mu + ig'B_\mu Y + igW^I_\mu T^I + ig_s G^A_\mu \mathscr{T}^A, \tag{1.4}$$

where Y is the weak hypercharge, $T^I = \tau^I/2$ are $SU(2)$ generators with τ^I being Pauli matrices and $\mathscr{T}^A = \lambda^A/2$ are $SU(3)$ generators with $\mathscr{T}^A$ being Gell-Mann matrices.
The field strength tensors read:

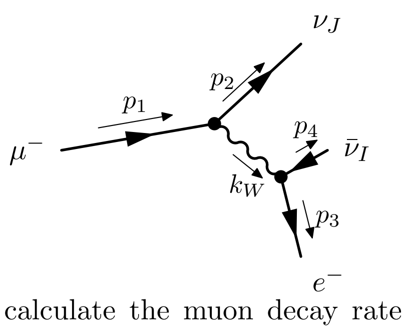
$$G^A_{\mu\nu} = \partial_\mu G^A_\nu - \partial_\nu G^A_\mu - g_s F^{ABC}G^B_\mu G^C_\nu, \tag{1.5}$$

$$W^I_{\mu\nu} = \partial_\mu W^A_\nu - \partial_\nu W^A_\mu - g\epsilon^{IJK}W^J_\mu W^K_\nu, \tag{1.6}$$

$$B_{\mu\nu} = \partial_\mu B_\nu - \partial_\nu B_\mu. \tag{1.7}$$

1.2. Muon decay in Standard Model

In Standard Model there is only one possible muon decay mode $\mu^- \to e^-\bar{\nu}_e\nu_\mu$:



(1.8)

Basing on this diagram, one may calculate the muon decay rate Γ_μ, muon lifetime τ_μ and Fermi constant G in Standard Model [3]:

$$\Gamma_\mu = \frac{m_\mu^5 G^2}{192\pi^3}, \quad \tau_\mu = \frac{192\pi^3}{m_\mu^5 G^2}, \quad G = \frac{\sqrt{2}g^2}{8m_W^2}, \tag{1.9}$$

where m_W is the mass of $W^\pm$ gauge boson and g is the weak interactions coupling constant.

Chapter 2

Muon decay in SMEFT

2.1. Relevant operators and mass basis

Process of muon decay in SMEFT is affected by additional operators, from which one may derive Feynman rules necessary for further calculations of scattering amplitudes. Below, we collect only those operators which gives non-zero contribution to muon decay. All possible dimension 6 operators were derived and presented in [4].

Table 2.1: SMEFT operators affecting the muon decay process

Q_{eW}	$(\bar{l}'_p \sigma^{\mu\nu} e'_r)\tau^I \varphi W^I_{\mu\nu}$	Q_{eB}	$(\bar{l}'_p \sigma^{\mu\nu} e'_r)\tau^I \varphi W^I_{\mu\nu}$
$Q_{\varphi WB}$	$\varphi^\dagger \tau^I \varphi W^I_{\mu\nu} B^{\mu\nu}$	$Q_{\varphi\Box}$	$(\varphi^\dagger\varphi)\Box(\varphi^\dagger\varphi)$
$Q_{\varphi D}$	$(\varphi^\dagger D^\mu \varphi)^*(\varphi^\dagger D_\mu \varphi)$	$Q_{e\varphi}$	$(\varphi^\dagger\varphi)(\bar{l}'_p e'_r \varphi)$
$Q^{(1)}_{\varphi l}$	$(\varphi^\dagger i \overleftrightarrow{D}_\mu \varphi)(\bar{l}'_p \gamma^\mu l'_r)$	$Q^{(3)}_{\varphi l}$	$(\varphi^\dagger i \overleftrightarrow{D}^I_\mu \varphi)(\bar{l}'_p \tau^I \gamma^\mu l'_r)$
$Q_{\varphi e}$	$(\varphi^\dagger i \overleftrightarrow{D}_\mu \varphi)(\bar{e}'_p \gamma^\mu e'_r)$	Q_{ll}	$(\bar{l}'_p \gamma_\mu l'_r)(\bar{l}'_s \gamma^\mu l'_t)$
Q_{ee}	$(\bar{e}'_p \gamma_\mu e'_r)(\bar{e}'_s \gamma^\mu e_t)$	Q_{le}	$(\bar{l}'_p \gamma_\mu l'_r)(\bar{e}'_s \gamma^\mu e_t)$

Mass eigenstates basis in SMEFT is derived in [1] and full procedure won't be repeated here. However it is proper to mention differences between Standard Model and SMEFT in this case.

Higgs potential is changed by additional operators and therefore, “new” vacuum expectation value (vev) reads:

$$v = \sqrt{\frac{2m^2}{\lambda}} + \frac{3m^2}{\sqrt{2}\lambda^{5/2}} C^{\varphi}. \tag{2.1}$$

In order to preserve canonical normalisation of kinetic terms in SMEFT one has to rescale fields and couplings in the following way.

- Scalar fields.

$$h = Z_h H, \quad G^0 = Z_{G^0} \Phi^0, \quad G^\pm \equiv \Phi^\pm, \tag{2.2}$$

where h is physical Higgs field and G^0, $G^\pm$ are Goldstone fields and where:

$$Z_h \equiv 1 + \frac{1}{4} C^{\varphi D} v^2 - C^{\varphi\Box} v^2, \tag{2.3}$$

$$Z_{G^0} \equiv 1 + \frac{1}{4} C^{\varphi D} v^2, \tag{2.4}$$

squared mass of Higgs field h reads then:

$$M_h^2 = \lambda v^2 - \left(3C^{\varphi} - 2\lambda C^{\varphi\Box} + \frac{\lambda}{2} C^{\varphi D} \right) v^4. \tag{2.5}$$

- Gauge fields and couplings.

$$\begin{aligned} \bar{W}^I_\mu &\equiv Z_g W^I_\mu, \quad \bar{g} = Z_g^{-1} g, \\ \bar{B}_\mu &\equiv Z_{g'} B_\mu, \quad \bar{g}' = Z_{g'}^{-1} g', \\ \bar{G}^A_\mu &\equiv Z_{g_s} G^A_\mu, \quad \bar{g}_s = Z_{g_s}^{-1} g_s, \end{aligned} \tag{2.6}$$

where:

$$\begin{aligned} Z_g &\equiv 1 - C^{\varphi W} v^2, \\ Z_{g'} &\equiv 1 - C^{\varphi B} v^2, \\ Z_{g_s} &\equiv 1 - C^{\varphi G} v^2. \end{aligned} \tag{2.7}$$

After those transformations, the form of covariant derivative is preserved:

$$D_\mu = \bar{D}_\mu = \partial_\mu + i\bar{g}' \bar{B}_\mu Y + i\bar{g} \bar{W}^I_\mu T^I + i\bar{g}_s \bar{G}^A_\mu \mathscr{T}^A. \tag{2.8}$$

Above redefinitions affect also gauge boson masses which now read:

$$M_W = \frac{1}{2} \bar{g} v, \tag{2.9}$$

$$M_Z = \frac{1}{2} \sqrt{\bar{g}^2 + \bar{g}'^2} v \left(1 + \frac{\epsilon \bar{g} \bar{g}'}{\bar{g}^2 + \bar{g}'^2}\right) Z_{G^0}, \tag{2.10}$$

$$M_A = 0, \tag{2.11}$$

where:

$$\epsilon \equiv C^{\varphi WB} v^2. \tag{2.12}$$

- Fermion fields.

  In order to diagonalize mass matrices, one has to rotate fermion fields by unitary matrices:

$$\Psi'_X = U_{\Psi_X} \Psi_X. \tag{2.13}$$

  In SM diagonalization of fermion mass matrices results in appearance of CKM and PMNS matrices which affect charged quark and lepton currents. SMEFT operators give additional contribution to those processes, with relevant part of Lagrangian for leptons:

$$\mathscr{L}_{c.c.} = -\frac{\bar{g}}{\sqrt{2}} W^+_\mu \bar{\nu}_i \gamma^\mu \left[U^\dagger_{e_L} (1 + v^2 C'^{\varphi l(3)}) U_{\nu_L}\right]^\dagger_{ij} P_L e_j + h.c., \tag{2.14}$$

  then mixing matrices are no longer unitary and for leptons becomes:

$$U_{PMNS} \equiv U \equiv U^\dagger_{e_L} (1 + v^2 C'^{\varphi l(3)}) U_{\nu_L}, \tag{2.15}$$

  with definition of $C^{\varphi l(3)}$ in mass basis:

$$C^{\varphi l(3)} = U^\dagger_{e_L} C'^{\varphi l(3)} U_{e_L}. \tag{2.16}$$

2.2. Feynman rules

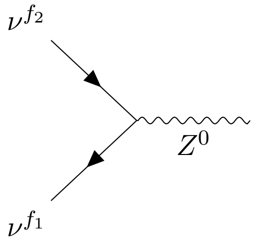
In this section, we present Feynman rules from [1], which one has to consider while dealing with muon decay process in SMEFT, derived from operators in Table 2.1.

2.2.1. General notation

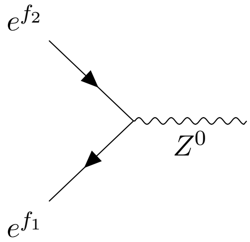
General conventions are adapted from Peskin-Schroeder book [5]. For all vertices, momenta are considered incoming. Minkowski metric tensor $\eta_{\mu\nu}$ signature reads (+,-,-,-). Below, we list additional symbols used to label fields:

Index type	Symbol
Flavour (generation)	f_i, g_i
Spinor	s_i
Lorentz	μ_i, ν_i, α_i...

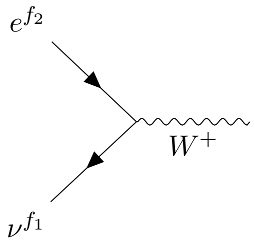
2.2.2. Vertices



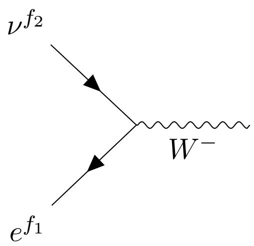
$$
\begin{aligned}
&= \frac{1}{2} i \sqrt{\bar{g}^2+\bar{g}'^2}\delta_{f_1f_2}\gamma^{\mu_3}\gamma^5 + \frac{i\bar{g}\bar{g}'v^2}{2\sqrt{\bar{g}^2+\bar{g}'^2}}\delta_{f_1f_2}\gamma^{\mu_3}\gamma^5 C^{\varphi WB} \\
&+ \frac{1}{2} i \sqrt{\bar{g}^2+\bar{g}'^2}\left[U_{g_2f_2}U^*_{g_1f_1}C^{\varphi l_1}_{g_1g_2}\gamma^{\mu_3}P_L - U_{g_1f_1}U^*_{g_2f_2}C^{\varphi l_1}_{g_2g_2}\gamma^{\mu_3}P_R\right] \\
&- \frac{1}{2} i \sqrt{\bar{g}^2+\bar{g}'^2}\left[U_{g_2f_2}U^*_{g_1f_1}C^{\varphi l_3}_{g_1g_2}\gamma^{\mu_3}P_L - U_{g_1f_1}U^*_{g_2f_2}C^{\varphi l_3}_{g_2g_2}\gamma^{\mu_3}P_R\right]
\end{aligned} \tag{2.17}
$$



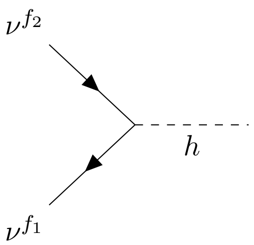
$$
\begin{aligned}
&-\frac{i}{2\sqrt{\bar{g}^2+\bar{g}'^2}}\delta_{f_1f_2}\left((\bar{g}'^2-\bar{g}^2)\gamma^{\mu_3}P_L - 2\bar{g}^2\gamma^{\mu_3}P_R\right) \\
&+\frac{i\bar{g}\bar{g}'v^2}{2(\bar{g}^2+\bar{g}'^2)^{3/2}}\delta_{f_1f_2}C^{\varphi WB}\left((\bar{g}'^2-\bar{g}^2)\gamma^{\mu_3}P_L + 2\bar{g}^2\gamma^{\mu_3}P_R\right) \\
&+\frac{\sqrt{2}v\bar{g}}{\sqrt{\bar{g}^2+\bar{g}'^2}}k^\nu_Z\left(C^{eW*}_{f_2f_1}\sigma^{\mu_3\nu}P_L + C^{eW}_{f_1f_2}\sigma^{\mu_3\nu}P_R\right) \\
&+\frac{\sqrt{2}v\bar{g}'}{\sqrt{\bar{g}^2+\bar{g}'^2}}k^\nu_Z\left(C^{eB*}_{f_2f_1}\sigma^{\mu_3\nu}P_L + C^{eB}_{f_1f_2}\sigma^{\mu_3\nu}P_R\right) \\
&+\frac{1}{2}iv^2\sqrt{\bar{g}^2+\bar{g}'^2}\gamma^{\mu_3}\left(C^{\varphi l_1}_{f_1f_2}P_L + C^{\varphi l_3}_{f_1f_2}P_L + C^{\varphi e}_{f_1f_2}P_R\right)
\end{aligned} \tag{2.18}
$$



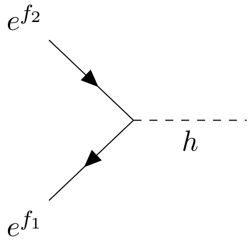
$$
-\frac{i\bar{g}}{\sqrt{2}}U^*_{f_2f_1}\gamma^{\mu_3}P_L - 2vk^\nu_W U^*_{g_1f_1}C^{eW}_{g_1f_2}\sigma^{\mu_3\nu}P_R \tag{2.19}
$$



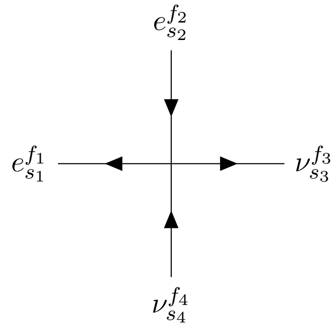
$$
-\frac{i\bar{g}}{\sqrt{2}}U_{f_1f_2}\gamma^{\mu_3}P_L - 2vk^\nu_W U_{g_2f_2}C^{eW*}_{f_1g_2}\sigma^{\mu_3\nu}P_L \tag{2.20}
$$



$$
\frac{2i}{v}\delta_{f_1f_2}m_{\nu_{f_1}} \tag{2.21}
$$



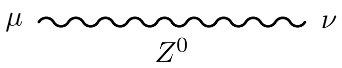
$$
\begin{aligned}
&-\frac{i}{v}\delta_{f_1f_2}m_{l_{f_1}} - iv\delta_{f_1f_2}C^{\varphi\Box}m_{l_{f_1}} + \frac{iv}{4}\delta_{f_1f_2}C^{\varphi D}m_{l_{f_1}} \\
&+\frac{iv^2}{\sqrt{2}}\left(P_L C^{e\varphi}_{f_2f_1} + P_R C^{e\varphi*}_{f_1f_2}\right)
\end{aligned} \tag{2.22}
$$



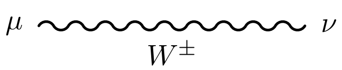
$$\begin{aligned}
&2i(\gamma^\mu P_L)_{s_1 s_2}\left[U_{g_2 f_4} U^*_{g_1 f_3}(\gamma^\mu P_L)_{s_3 s_4} C^{ll}_{f_1 f_2 g_1 g_2}\right] \\
&-2i(\gamma^\mu P_L)_{s_1 s_2}\left[U_{g_1 f_3} U^*_{g_2 f_4}(\gamma^\mu P_R)_{s_3 s_4} C^{ll}_{f_1 f_2 g_2 g_1}\right] \\
&+i(\gamma^\mu P_R)_{s_1 s_2}\left[U_{g_2 f_4} U^*_{g_1 f_3}(\gamma^\mu P_L)_{s_3 s_4} C^{ll}_{g_1 g_2 f_1 f_2}\right] \\
&-i(\gamma^\mu P_R)_{s_1 s_2}\left[U_{g_1 f_3} U^*_{g_2 f_4}(\gamma^\mu P_R)_{s_3 s_4} C^{ll}_{g_2 g_1 f_1 f_2}\right]
\end{aligned} \tag{2.23}$$

2.2.3. Propagators in Unitary gauge

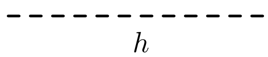
All calculations are performed at the tree-level, so in this case choice of unitary gauge is sufficient. In unitary gauge, necessary propagators reads:



$$-\frac{i}{k_Z^2 - M_Z^2}\left(\eta^{\mu\nu} - \frac{k_Z^\mu k_Z^\nu}{k_Z^2 - M_Z^2}\right) \tag{2.24}$$

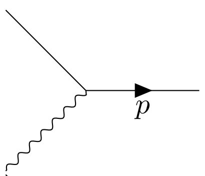


$$-\frac{i}{k_W^2 - M_W^2}\left(\eta^{\mu\nu} - \frac{k_W^\mu k_W^\nu}{k_W^2 - M_W^2}\right) \tag{2.25}$$

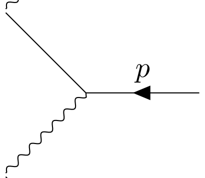


$$\frac{i}{k_h^2 - M_h^2} \tag{2.26}$$

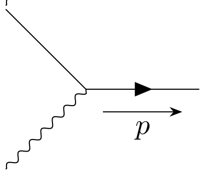
2.2.4. External fermions and anti-fermions



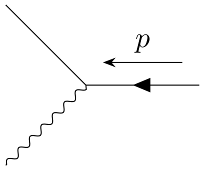
$$= u(p,s) \text{ (initial)} \tag{2.27}$$



$$= \bar{u}(p,s) \text{ (final)} \tag{2.28}$$



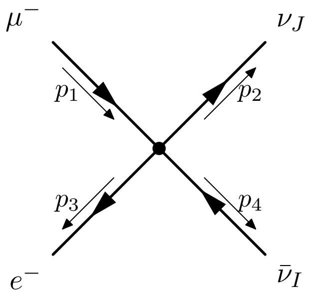
$$= \bar{v}(p,s) \text{ (initial)} \tag{2.29}$$



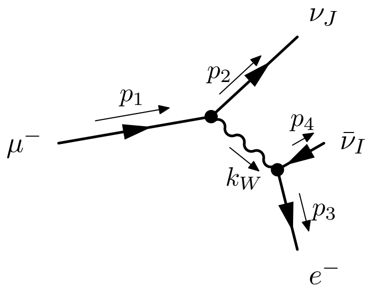
$= v(p, s)$ (final) (2.30)

2.3. Diagrams

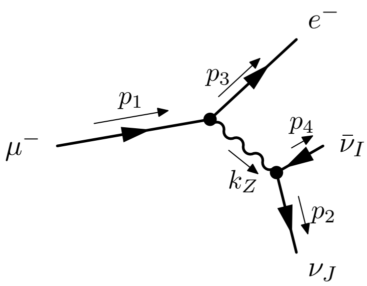
In this section, we collect all diagrams contributing to the muon decay amplitude in SMEFT:

4-fermion diagram = 

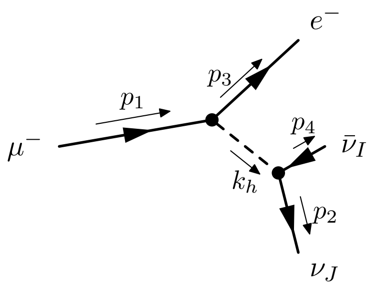
momentum conservation: $p_1 = p_2 + p_3 + p_4$ (2.31)

$W^{\pm}$-propagator diagram = 

momentum conservation: $k_W = p_1 - p_2 = p_3 + p_4$ (2.32)

Z^0-propagator diagram = 

momentum conservation: $k_Z = p_1 - p_3 = p_2 + p_4$ (2.33)

h -propagator diagram = 

momentum conservation: $k_h = p_1 - p_3 = p_2 + p_4$ (2.34)

Chapter 3

Methodology and calculations

3.1. Methodology

In order to calculate the decay rate of muon in SMEFT, one has to perform the following calculations:

1. calculate total amplitude $i\mathscr{M}_{tot}$ as a sum of amplitudes of processes 2.31, 2.32, 2.33, 2.34 using Feynman rules presented in section 2.2;
2. calculate squared amplitude, average over initial spins s_i and sum over final spins s_f:
$$|\bar{\mathscr{M}}|^2 = \frac{1}{2}\sum_{s_{fi}} \mathscr{M}_{tot}\mathscr{M}^\dagger_{tot};$$
3. using Fermi's golden rule, given by equation (A.35), calculate the decay rate Γ of muon in SMEFT.

The necessary standard gamma matrix identities, Dirac equation conventions and phase space expressions are collected in Appendix A.

3.2. Decay amplitude

Momenta are assigned to particles as on diagrams 2.31, 2.32, 2.33, 2.34:

$$p_\mu = p_1, \quad p_e = p_3, \quad p_{\nu_J} = p_2, \quad p_{\bar{\nu}_I} = p_4. \tag{3.1}$$

It is assumed, that only muon mass m_μ is non negligible:

$$\begin{gathered} m_e = m_{\nu_J} = m_{\bar{\nu}_I} \approx 0, \\ m_\mu \neq 0. \end{gathered} \tag{3.2}$$

Following this assumption, Dirac equation (A.31, A.32) for fermions can be rewritten in the form:

$$\begin{gathered} \not{p}_1 u(p_1) = m_\mu u(p_1), \\ \bar{u}(p_1)\not{p}_1 = \bar{u}(p_1) m_\mu, \end{gathered} \tag{3.3}$$

$$\begin{gathered} \not{p}_{2,3} u(p_{2,3}) = 0, \\ \bar{u}(p_{2,3})\not{p}_{2,3} = 0, \end{gathered} \tag{3.4}$$

$$\begin{gathered} \bar{v}(p_4)\not{p}_4 = 0, \\ \not{p}_4 v(p_4) = 0. \end{gathered} \tag{3.5}$$

Similarly, polarisation sum formulae (A.33, A.34) read:

$$\sum_{s_1} u(p_1, s_1)\bar{u}(p_1, s_1) = \not{p}_1 + m_\mu, \tag{3.6}$$

$$\sum_{s_{2,3}} u(p_{2,3}, s_{2,3})\bar{u}(p_{2,3}, s_{2,3}) = \not{p}_{2,3}, \tag{3.7}$$

$$\sum_{s_4} v(p_4, s_4)\bar{v}(p_4, s_4) = \not{p}_4. \tag{3.8}$$

3.2.1. 4-fermion diagram

For diagram 2.31 one needs to use Feynman rule given by eq (2.23). Amplitude then reads:

$$\begin{aligned}
i\mathscr{M}_{4-fermion} = {} & 2iU_{g_4 I}U^*_{g_3 J}C^{ll}_{12g_3g_4}\bar{u}_e(p_3)\gamma^\mu P_L u_\mu(p_1)\bar{u}_{\nu_J}(p_2)\gamma_\mu P_L v_{\bar{\nu}_I}(p_4) \\
& -2iU_{g_3 J}U^*_{g_4 I}C^{ll}_{12g_4g_3}\bar{u}_e(p_3)\gamma^\mu P_L u_\mu(p_1)\bar{u}_{\nu_J}(p_2)\gamma_\mu P_R v_{\bar{\nu}_I}(p_4) \\
& +iU_{g_4 I}U^*_{g_3 J}C^{le}_{g_3g_4 12}\bar{u}_e(p_3)\gamma^\mu P_R u_\mu(p_1)\bar{u}_{\nu_J}(p_2)\gamma_\mu P_L v_{\bar{\nu}_I}(p_4) \\
& -iU_{g_3 J}U^*_{g_4 I}C^{l3}_{g_4g_3 12}F_4\bar{u}_e(p_3)\gamma^\mu P_R u_\mu(p_1)\bar{u}_{\nu_J}(p_2)\gamma_\mu P_R v_{\bar{\nu}_I}(p_4).
\end{aligned} \tag{3.9}$$

3.2.2. $W^\pm$ propagator diagram

For diagram 2.32 amplitude reads:

$$i\mathscr{M}_W = \bar{u}_{\nu_J}(p_2)V_1^\mu u_\mu(p_1)\frac{-i}{k_W^2 - M_W^2}\left(\eta_{\mu\nu} - \frac{k_{W\mu}k_{W\nu}}{k_W^2 - M_W^2}\right)\bar{u}_e(p_3)V_2^\nu v_{\bar{\nu}I}(p_4), \tag{3.10}$$

assuming that $M_W^2 >> k_W^2 \sim m_\mu^2$ one gets:

$$\begin{aligned}
i\mathscr{M}_W = {} & \frac{i}{M_W^2}\bar{u}_{\nu_J}(p_2)V_1^\mu u_\mu(p_1)\bar{u}_e(p_3)V_{2\mu}v_{\bar{\nu}I}(p_4) \\
& -\frac{i}{M_W^4}\bar{u}_{\nu_J}(p_2)V_1^\mu k_{W\mu}u_\mu(p_1)\bar{u}_e(p_3)V_2^\nu k_{W\nu}v_{\bar{\nu}I}(p_4).
\end{aligned} \tag{3.11}$$

Vertex factors from eqs. (2.19, 2.20) are:

$$V_1^\mu = -\frac{i\bar{g}}{\sqrt{2}}U^*_{2J}\gamma^\mu P_L - 2vk_{W\beta}U^*_{g_1 J}C^{eW}_{g_1 2}\sigma^{\mu\beta}P_R, \tag{3.12}$$

$$V_2^\nu = -\frac{i\bar{g}}{\sqrt{2}}U_{1I}\gamma^\nu P_L - 2vk_{W\alpha}U_{g_2 I}C^{eW*}_{1g_2}\sigma^{\nu\alpha}P_L. \tag{3.13}$$

For clarity of calculations, it is convenient to introduce some symbols:

$$V_{1a}^\mu = -\frac{i\bar{g}}{\sqrt{2}}U^*_{2J}\gamma^\mu P_L, \tag{3.14}$$

$$V_{1b}^\mu = -2vk_{W\beta}U^*_{g_1 J}C^{eW}_{g_1 2}\sigma^{\mu\beta}P_R, \tag{3.15}$$

$$V_{2a}^\nu = -\frac{i\bar{g}}{\sqrt{2}}U_{1I}\gamma^\nu P_L, \tag{3.16}$$

$$V_{2b}^\nu = -2vk_{W\alpha}U_{g_2 I}C^{eW*}_{1g_2}\sigma^{\nu\alpha}P_L. \tag{3.17}$$

It is appropriate to treat both expressions in eq. (3.11) separately:

$$i\mathscr{M}_{W1} = \frac{i}{M_W^2}\bar{u}_{\nu_J}(p_2)V_1^\mu u_\mu(p_1)\bar{u}_e(p_3)V_{2\mu}v_{\bar{\nu}_I}(p_4), \tag{3.18}$$

$$i\mathscr{M}_{W2} = -\frac{i}{M_W^4}\bar{u}_{\nu_J}(p_2)V_1^\mu k_{W\mu}u_\mu(p_1)\bar{u}_e(p_3)V_2^\nu k_{W\nu}v_{\bar{\nu}I}(p_4). \tag{3.19}$$

First part of the amplitude, obtained from inserting eqs. (3.14, 3.16) to eq. (3.18) reads:

$$i\mathscr{M}_{W1aa} = D_1\bar{u}_{\nu_J}(p_2)\gamma^\mu P_L u_\mu(p_1)\bar{u}_e(p_3)\gamma_\mu P_L v_{\bar{\nu}I}(p_4), \tag{3.20}$$

where:

$$D_1 = \frac{-i\bar{g}^2U^*_{2J}U_{1I}}{2M_W^2}.$$

Second part of the amplitude from eqs. (3.14, 3.17) with eq. (3.18):

$$\begin{aligned}
i\mathscr{M}_{W1ab} = {} & D_2\bar{u}_{\nu_J}(p_2)\gamma^\mu P_L u_\mu(p_1)\bar{u}_e(p_3)k_W^\alpha\sigma_{\mu\alpha}P_L v_{\bar{\nu}I}(p_4) = \\
& D_2'\bar{u}_{\nu_J}(p_2)\gamma^\mu P_L u_\mu(p_1)\bar{u}_e(p_3)k_W^\alpha(\gamma_\mu\gamma_\alpha - \gamma_\alpha\gamma_\mu)P_L v_{\bar{\nu}I}(p_4) \\
= {} & D_2'\bar{u}_{\nu_J}(p_2)\gamma^\mu P_L u_\mu(p_1)\bar{u}_e(p_3)(\gamma_\mu(\not{p}_3 + \not{p}_4) - (\not{p}_3 + \not{p}_4)\gamma_\mu)P_L v_{\bar{\nu}I}(p_4),
\end{aligned}$$

using eqs. (3.5, 3.4, A.3):

$$\begin{aligned}
i\mathscr{M}_{W_1ab} &= D_2'\bar{u}_{\nu_J}(p_2)\gamma^\mu P_L u_\mu(p_1)\bar{u}_e(p_3)(\gamma_\mu \not{p_3} - \not{p_4}\gamma_\mu)P_L v_{\bar{\nu}I}(p_4)\\
&= D_2'\bar{u}_{\nu_J}(p_2)\gamma^\mu P_L u_\mu(p_1)\bar{u}_e(p_3)(2\eta_{\mu\alpha}(p_3-p_4)^\alpha + \gamma_\mu \not{p_4} - \not{p_3}\gamma_\mu)P_L v_{\bar{\nu}I}(p_4) =\\
&= 2D_2'\bar{u}_{\nu_J}(p_2)(\not{p_3} - \not{p_4})P_L u_\mu(p_1)\bar{u}_e(p_3)P_L v_{\bar{\nu}I}(p_4),
\end{aligned}$$

using momentum conservation formula: $p_4 = p_1 - p_2 - p_3$ and eqs. (3.4, 3.3) one gets:

$$\begin{aligned}
i\mathscr{M}_{W_1ab} &= -2m_\mu D_2'\bar{u}_{\nu_J}(p_2)P_R u_\mu(p_1)\bar{u}_e(p_3)P_L v_{\bar{\nu}I}(p_4)\\
&+4D_2'\bar{u}_{\nu_J}(p_2)\not{p_3}P_L u_\mu(p_1)\bar{u}_e(p_3)P_L v_{\bar{\nu}I}(p_4),
\end{aligned}$$

after renaming constants the result is given by:

$$\begin{aligned}
i\mathscr{M}_{W_1ab} &= -D_{22}\bar{u}_{\nu_J}(p_2)P_R u_\mu(p_1)\bar{u}_e(p_3)P_L v_{\bar{\nu}I}(p_4)\\
&+D_{21}\bar{u}_{\nu_J}(p_2)\not{p_3}P_L u_\mu(p_1)\bar{u}_e(p_3)P_L v_{\bar{\nu}I}(p_4),
\end{aligned} \tag{3.21}$$

where:

$$D_{21} = \frac{-i2\sqrt{2}v\bar{g}U^*_{2J}U_{g_2I}C^{eW*}_{1g_2}}{M_W^2} \quad D_{22} = \frac{-i\sqrt{2}vm_\mu\bar{g}U^*_{2J}U_{g_2I}C^{eW*}_{1g_2}}{M_W^2}.$$

Next part of the amplitude from eqs. (3.15, 3.16, 3.18):

$$i\mathscr{M}_{W_1ba} = D_3\bar{u}_{\nu_J}(p_2)k_{W_\beta}\sigma^{\mu\beta}P_R u_\mu(p_1)\bar{u}_e(p_3)\gamma_m u P_L v_{\bar{\nu}I}(p_4),$$

calculations are analogous to previous ones:

$$\begin{aligned}
i\mathscr{M}_{W_1ba} &= D_3'\bar{u}_{\nu_J}(p_2)k_{W_\beta}(\gamma^\mu\gamma^\beta - \gamma^\beta\gamma^\mu)P_R u_\mu(p_1)\bar{u}_e(p_3)\gamma_\mu P_L v_{\bar{\nu}I}(p_4) =\\
&D_3'\bar{u}_{\nu_J}(p_2)(\gamma^\mu(\not{p_1} - \not{p_2}) - (\not{p_1} - \not{p_2})\gamma^\mu)P_R u_\mu(p_1)\bar{u}_e(p_3)\gamma_\mu P_L v_{\bar{\nu}I}(p_4),
\end{aligned}$$

using eqs. (A.3, 3.3, 3.4):

$$\begin{aligned}
i\mathscr{M}_{W_1ba} &= 2D_3'm_\mu\bar{u}_{\nu_J}(p_2)\gamma^\mu P_L u_\mu(p_1)\bar{u}_e(p_3)\gamma_\mu P_L v_{\bar{\nu}I}(p_4)+\\
&-2D_3'\bar{u}_{\nu_J}(p_2)P_R u_\mu(p_1)\bar{u}_e(p_3)(\not{p_1} + \not{p_2})P_L v_{\bar{\nu}I}(p_4),
\end{aligned}$$

using momentum conservation $p_1 = p_2 + p_3 + p_4$, eqs. (3.4,3.5) one obtains the final result:

$$\begin{aligned}
i\mathscr{M}_{W_1ba} &= D_{31}\bar{u}_{\nu_J}(p_2)\gamma^\mu P_L u_\mu(p_1)\bar{u}_e(p_3)\gamma_\mu P_L v_{\bar{\nu}I}(p_4)+\\
&-D_{32}\bar{u}_{\nu_J}(p_2)P_R u_\mu(p_1)\bar{u}_e(p_3)\not{p_2}P_L v_{\bar{\nu}I}(p_4),
\end{aligned} \tag{3.22}$$

where:

$$D_{31} = \frac{i\sqrt{2}vm_\mu\bar{g}U^*_{g_1J}U_{1I}C^{eW}_{g_12}}{M_W^2} \quad D_{32} = \frac{i2\sqrt{2}v\bar{g}U^*_{g_1J}U_{1I}C^{eW}_{g_12}}{M_W^2}.$$

Subsequent part of the amplitude from eqs. (3.15, 3.17, 3.18) has form:

$$iM_{W_1bb} = D_4\bar{u}_{\nu_J}(p_2)k_{W\beta}\sigma^{\mu\beta}P_R u_\mu(p_1)\bar{u}_e(p_3)k_W^\alpha\sigma_{\mu\alpha}P_L v_{\bar{\nu}I}(p_4),$$

where D_4 written explicitly reads:

$$D_4 = -4v^2U^*_{g_1J}C^{eW}_{g_12}U_{g_2I}C^{eW*}_{1g_2},$$

one can now argue that, since D_4 is proportional to C^2, whole term $i\mathscr{M}_{W_1bb} \propto \frac{1}{\Lambda^4}$ and therefore:

$$i\mathscr{M}_{W_1bb} = 0. \tag{3.23}$$

One is left with the part of the amplitude given by eq. (3.19). It is sufficient to concentrate only on the second part of this expression which reads:

$$A = \bar{u}_e(p_3)V_2^\nu k_{W\nu}v_{\bar{\nu}I}(p_4), \tag{3.24}$$

first part, obtained from eq. (3.16) has following form:

$$\begin{aligned}A_a &= D_5 \bar{u}_e(p_3)\gamma^\nu P_L k_{W\nu} v_{\bar{\nu}I}(p_4) = \\ &= D_5 \bar{u}_e(p_3) P_R(\not{p_3} + \not{p_4}) v_{\bar{\nu}I}(p_4),\end{aligned}$$

applying eqs. (3.4,3.5) this part vanishes:

$$A_a = 0,$$

from eq. (3.17) next term reads:

$$\begin{aligned}A_b &= D_6 \bar{u}_e(p_3) k_{W\alpha}\sigma^{\nu\alpha} P_L k_{W\nu}(p_3+p_4) v_{\bar{\nu}I}(p_4) = \\ &= D_5' \bar{u}_e(p_3)[(\not{p_3}+\not{p_4})(\not{p_3}+\not{p_4}) - (\not{p_3}+\not{p_4})(\not{p_3}+\not{p_4})] P_L v_{\bar{\nu}I}(p_4) = 0,\end{aligned}$$

and whole expression:

$$A = \bar{u}_e(p_3) V_2^\nu k_{W\nu} v_{\bar{\nu}I}(p_4) = 0,$$

therefore, since:

$$i\mathscr{M}_{W2} = -\frac{i}{M_W^2}\bar{u}_{\nu_J}(p_2) V_1^\mu u_\mu(p_1) A,$$

one may write down the result:

$$i\mathscr{M}_{W2} = 0. \tag{3.25}$$

Finally, the whole expression for $W^\pm$ diagram amplitude is given by:

$$\begin{aligned}i\mathscr{M}_W = i\mathscr{M}_{W1} = (D_1 + D_{31})\bar{u}_{\nu_J}(p_2)\gamma^\mu P_L u_\mu(p_1)\bar{u}_e(p_3)\gamma_\mu P_L v_{\bar{\nu}I}(p_4) \\ -D_{22}\bar{u}_{\nu_J}(p_2) P_R u_\mu(p_1)\bar{u}_e(p_3) P_L v_{\bar{\nu}I}(p_4) \\ +D_{21}\bar{u}_{\nu_J}(p_2)\not{p_3} P_L u_\mu(p_1)\bar{u}_e(p_3) P_L v_{\bar{\nu}I}(p_4) \\ -D_{32}\bar{u}_{\nu_J}(p_2) P_R u_\mu(p_1)\bar{u}_e(p_3)\not{p_2} P_L v_{\bar{\nu}I}(p_4).\end{aligned} \tag{3.26}$$

3.2.3. Z^0 propagator diagram

For diagram 2.33 the amplitude reads:

$$i\mathscr{M}_Z = \bar{u}_e(p_3) V_1^\mu u_\mu(p_1)\frac{-i}{k_Z^2 - M_Z^2}\left(\eta_{\mu\nu} - \frac{k_{Z\mu}k_{Z\nu}}{k_Z^2 - M_Z^2}\right)\bar{u}_{\nu_J}(p_2)(p_3) V_2^\nu v_{\bar{\nu}I}(p_4), \tag{3.27}$$

assuming again that $M_Z^2 >> k_Z^2 \sim m_\mu^2$ one gets:

$$\begin{aligned}i\mathscr{M}_Z = \frac{i}{M_Z^2}\bar{u}_e(p_3) V_1^\mu u_\mu(p_1)\bar{u}_{\nu_J}(p_2) V_{2\mu} v_{\bar{\nu}I}(p_4) \\ -\frac{i}{M_Z^4}\bar{u}_e(p_e) V_1^\mu k_{Z\mu} u_\mu(p_1)\bar{u}_{\nu_J}(p_2) V_2^\nu k_{W\nu} v_{\bar{\nu}I}(p_4).\end{aligned} \tag{3.28}$$

Vertex factors from Feynman rules given by eqs. (2.17, 2.18) read:

$$\begin{aligned}V_1^\mu = -\frac{i}{2\sqrt{\bar{g}^2+\bar{g}'^2}}\delta_{12}\left((\bar{g}'^2 - \bar{g}^2)\gamma^{\mu_3} P_L - 2\bar{g}^2\gamma^{\mu_3} P_R\right) \\ +\frac{i\bar{g}\bar{g}'v^2}{2(\bar{g}^2+\bar{g}'^2)^{3/2}}\delta_{12} C^{\varphi WB}\left((\bar{g}'^2 - \bar{g}^2)\gamma^{\mu_3} P_L + 2\bar{g}^2\gamma^{\mu_3} P_R\right) \\ +\frac{\sqrt{2}v\bar{g}}{\sqrt{\bar{g}^2+\bar{g}'^2}} k_Z^\nu\left(C_{21}^{eW*}\sigma^{\mu_3\nu} P_L + C_{12}^{eW}\sigma^{\mu_3\nu} P_R\right) \\ +\frac{\sqrt{2}v\bar{g}'}{\sqrt{\bar{g}^2+\bar{g}'^2}} k_Z^\nu\left(C_{21}^{eB*}\sigma^{\mu_3\nu} P_L + C_{12}^{eB}\sigma^{\mu_3\nu} P_R\right) \\ +\frac{1}{2}iv^2\sqrt{\bar{g}^2+\bar{g}'^2}\gamma^{\mu_3}\left(C_{12}^{\varphi l_1} P_L + C_{12}^{\varphi l_3} P_L + C_{12}^{\varphi e} P_R\right),\end{aligned} \tag{3.29}$$

$$
\begin{aligned}
V_2^{\nu} = \frac{1}{2} i\sqrt{\bar{g}^2+\bar{g}'^2}\delta_{JI}\gamma^{\mu_3}\gamma^5 + \frac{i\bar{g}\bar{g}'v^2}{2\sqrt{\bar{g}^2+\bar{g}'^2}}\delta_{JI}\gamma^{\mu_3}\gamma^5 C^{\varphi WB} \\
+\frac{1}{2} i\sqrt{\bar{g}^2+\bar{g}'^2}\left(U_{g_2J}U^*_{g_1I}C^{\varphi l_1}_{g_1g_2}\gamma^{\mu_3}P_L - U_{g_1I}U^*_{g_2J}C^{\varphi l_1}_{g_2g_2}\gamma^{\mu_3}P_R\right) \\
-\frac{1}{2} i\sqrt{\bar{g}^2+\bar{g}'^2}\left(U_{g_2J}U^*_{g_1I}C^{\varphi l_3}_{g_1g_2}\gamma^{\mu_3}P_L - U_{g_1I}U^*_{g_2J}C^{\varphi l_3}_{g_2g_2}\gamma^{\mu_3}P_R\right).
\end{aligned}
\tag{3.30}
$$

Before moving on to calculations, one may observe some major simplifications in this part of amplitude:

1. in expression (3.29), since $\delta_{12}=0$, both terms containing δ_{12} will vanish;
2. as a result, all terms in eq. (3.29) will be proportional to Wilson coefficients C, and since $C^2 \propto \frac{1}{\Lambda^4} \to 0$, all terms in eq. (3.30) proportional to C will be neglected.

Finally one has to consider only the following terms:

$$
\begin{aligned}
V_1^{\mu} = +\frac{\sqrt{2}v\bar{g}}{\sqrt{\bar{g}^2+\bar{g}'^2}}k_Z^{\nu}\left(C^{eW*}_{21}\sigma^{\mu_3\nu}P_L + C^{eW}_{12}\sigma^{\mu_3\nu}P_R\right) \\
+\frac{\sqrt{2}v\bar{g}'}{\sqrt{\bar{g}^2+\bar{g}'^2}}k_Z^{\nu}\left(C^{eB*}_{21}\sigma^{\mu_3\nu}P_L + C^{eB}_{12}\sigma^{\mu_3\nu}P_R\right) \\
+\frac{1}{2} iv^2\sqrt{\bar{g}^2+\bar{g}'^2}\gamma^{\mu_3}\left(C^{\varphi l_1}_{12}P_L + C^{\varphi l_3}_{12}P_L + C^{\varphi e}_{12}P_R\right),
\end{aligned}
\tag{3.31}
$$

$$
V_2^{\nu} = \frac{1}{2} i\sqrt{\bar{g}^2+\bar{g}'^2}\delta_{JI}\gamma^{\mu_3}\gamma^5. \tag{3.32}
$$

Separating eq. (3.28) into two expressions and introducing symbols to eq. (3.31):

$$
i\mathscr{M}_{Z1} = \frac{i}{M_Z^2}\bar{u}_e(p_3)V_1^{\mu}u_{\mu}(p_1)\bar{u}_{\nu_J}(p_2)V_{2\mu}v_{\bar{\nu}I}(p_4), \tag{3.33}
$$

$$
i\mathscr{M}_{Z2} = -\frac{i}{M_Z^4}\bar{u}_e(p_3)V_1^{\mu}k_{Z\mu}u_{\mu}(p_1)\bar{u}_{\nu_J}(p_2)V_2^{\nu}k_{Z\nu}v_{\bar{\nu}I}(p_4), \tag{3.34}
$$

$$
V_{1a}^{\mu} = +\frac{\sqrt{2}v\bar{g}}{\sqrt{\bar{g}^2+\bar{g}'^2}}k_{Z\nu}\left(C^{eW*}_{21}\sigma^{\mu_3\nu}P_L + C^{eW}_{12}\sigma^{\mu_3\nu}P_R\right), \tag{3.35}
$$

$$
V_{1b}^{\mu} = +\frac{\sqrt{2}v\bar{g}'}{\sqrt{\bar{g}^2+\bar{g}'^2}}k_{Z\nu}\left(C^{eB*}_{21}\sigma^{\mu_3\nu}P_L + C^{eB}_{12}\sigma^{\mu_3\nu}P_R\right), \tag{3.36}
$$

$$
V_{1c}^{\mu} = +\frac{1}{2} iv^2\sqrt{\bar{g}^2+\bar{g}'^2}\gamma^{\mu_3}\left(C^{\varphi l_1}_{12}P_L + C^{\varphi l_3}_{12}P_L + C^{\varphi e}_{12}P_R\right). \tag{3.37}
$$

First part of the amplitude is obtained from eqs. (3.33, 3.35, 3.32):

$$
i\mathscr{M}_{Z1a} = B_1'\bar{u}_e(p_3)k_{Z\nu}\left(C^{eW*}_{21}\sigma^{\mu\nu}P_L + C^{eW}_{12}\sigma^{\mu\nu}P_R\right)u_{\mu}(p_1)\bar{u}_{\nu_J}(p_2)\gamma_{\mu}\gamma^5 v_{\bar{\nu}I}(p_4),
$$

using momentum conservation formula $k_Z = p_1 - p_3 = p_2 + p_4$ leads to:

$$
i\mathscr{M}_{Z1a} = B_1\bar{u}_e(p_3)(\gamma^{\mu}(\not{p}_1 - \not{p}_3) - (\not{p}_1 - \not{p}_3)\gamma^{\mu})\left(C^{eW*}_{21}P_L + C^{eW}_{12}P_R\right)u_{\mu}(p_1)\bar{u}_{\nu_J}(p_2)\gamma_{\mu}\gamma^5 v_{\bar{\nu}I}(p_4),
$$

using eqs.(3.3, 3.4) and eq. (A.3) allows to write:

$$
\begin{aligned}
i\mathscr{M}_{Z1a} = 2B_1 m_{\mu}\bar{u}_e(p_3)\gamma^{\mu}\left(C^{eW*}_{21}P_R + C^{eW}_{12}P_L\right)u_{\mu}(p_1)\bar{u}_{\nu_J}(p_2)\gamma_{\mu}\gamma^5 v_{\bar{\nu}I}(p_4) \\
-2B_1\bar{u}_e(p_3)\left(C^{eW*}_{21}P_L + C^{eW}_{12}P_R\right)u_{\mu}(p_1)\bar{u}_{\nu_J}(p_2)(\not{p}_3 + \not{p}_1)\gamma^5 v_{\bar{\nu}I}(p_4),
\end{aligned}
$$

applying eqs. (3.4, 3.5) and $p_1 = p_2 + p_3 + p_4$ one obtains the final result:

$$
\begin{aligned}
i\mathscr{M}_{Z1a} = 2B_1 m_{\mu}\bar{u}_e(p_3)\gamma^{\mu}\left(C^{eW*}_{21}P_R + C^{eW}_{12}P_L\right)u_{\mu}(p_1)\bar{u}_{\nu_J}(p_2)\gamma_{\mu}\gamma^5 v_{\bar{\nu}I}(p_4) \\
-4B_1\bar{u}_e(p_3)\left(C^{eW*}_{21}P_L + C^{eW}_{12}P_R\right)u_{\mu}(p_1)\bar{u}_{\nu_J}(p_2)\not{p}_3\gamma^5 v_{\bar{\nu}I}(p_4),
\end{aligned}
\tag{3.38}
$$

where:

$$B_1 = \frac{i\sqrt{2}\bar{g}v\delta_{IJ}}{4M_Z^2}.$$

$i\mathscr{M}_{Z1b}$, from eqs. (3.33, 3.36,3.32) is very similar to $i\mathscr{M}_{Z1a}$, the only difference are constants, so one can immediately write analogous expression:

$$\begin{aligned} i\mathscr{M}_{Z1b} = 2B_2 m_\mu \bar{u}_e(p_3)\gamma^\mu \left(C_{21}^{eB*}P_R + C_{12}^{eB}P_L\right) u_\mu(p_1)\bar{u}_{\nu_J}(p_2)\gamma_\mu\gamma^5 v_{\bar{\nu}I}(p_4) \\ -4B_2\bar{u}_e(p_3)\left(C_{21}^{eB*}P_L + C_{12}^{eB}P_R\right) u_\mu(p_1)\bar{u}_{\nu_J}(p_2)\not{p_3}\gamma^5 v_{\bar{\nu}I}(p_4), \end{aligned} \tag{3.39}$$

where:

$$B_2 = \frac{i\sqrt{2}\bar{g'}v\delta_{IJ}}{4M_Z^2}.$$

Last term, $i\mathscr{M}_{Z1c}$, from eqs. (3.33, 3.37, 3.32) cannot be simplified and reads:

$$i\mathscr{M}_{Z1c} = B_3\bar{u}_e(p_3)\left(\gamma^\mu P_L(C_{12}^{\varphi l_1} + C_{12}^{\varphi l_3}) + \gamma^\mu P_R C_{12}^{\varphi e}\right) u_\mu(p_1)\bar{u}_{\nu_J}(p_2)\gamma_\mu\gamma^5 v_{\bar{\nu}I}(p_4), \tag{3.40}$$

where:

$$B_3 = \frac{-i(\bar{g}^2 + \bar{g'}^2)v^2\delta_{IJ}}{4M_Z^2}.$$

One needs to calculate the part of amplitude given by expression (3.34).
At first, one can look closer only at the second part of this expression, taking $k_Z = p_2 + p_4$:

$$i\mathscr{M}_{Z2} = B_4\bar{u}_{e(p_3)}V_1^\mu k_{Z\mu}u_\mu(p_1)\bar{u}_{\nu_J}(p_2)(\not{p_2} + \not{p_4})\gamma^5 v_{\bar{\nu}I}(p_4),$$

which, after applying eqs. (3.4, 3.5), gives zero for every $V_{1a,b,c}^\mu$, therefore:

$$i\mathscr{M}_{Z2} = 0. \tag{3.41}$$

Finally the amplitude for Z-propagator diagram is given by the following formula:

$$\begin{aligned} i\mathscr{M}_Z = \phi_1\bar{u}_e(p_3)\gamma^\mu P_L u_\mu(p_1)\bar{u}_{\nu_J}(p_2)\gamma_\mu\gamma^5 v_{\bar{\nu}I}(p_4) \\ \phi_2\bar{u}_e(p_3)\gamma^\mu P_R u_\mu(p_1)\bar{u}_{\nu_J}(p_2)\gamma_\mu\gamma^5 v_{\bar{\nu}I}(p_4) \\ \phi_3\bar{u}_e(p_3)P_R u_\mu(p_1)\bar{u}_{\nu_J}(p_2)\not{p_3}\gamma^5 v_{\bar{\nu}I}(p_4) \\ \phi_4\bar{u}_e(p_3)P_L u_\mu(p_1)\bar{u}_{\nu_J}(p_2)\not{p_3}\gamma^5 v_{\bar{\nu}I}(p_4), \end{aligned} \tag{3.42}$$

where:

$$\begin{aligned} \phi_1 &= B_3(C_{12}^{\varphi l_1} + C_{12}^{\varphi l_3}) + 2B_1 m_\mu C_{12}^{eW} + 2B_2 m_\mu C_{12}^{eB} \\ \phi_2 &= B_3 C_{12}^{\varphi l_3} + 2B_1 m_\mu C_{21}^{eW*} + 2B_2 m_\mu C_{21}^{eB*} \\ \phi_3 &= -4(B_1 C_{12}^{eW} + B_2 C_{12}^{eB}) \\ \phi_4 &= -4(B_1 C_{21}^{eW*} + B_2 C_{21}^{eB*}). \end{aligned}$$

3.2.4. h propagator diagram

For diagram 2.34 the amplitude reads:

$$i\mathscr{M}_h = \bar{u}(p_3)V_1^\mu u(p_1)\frac{i}{k_h^2 - M_h^2}\bar{u}(p_2)V_2^\nu v(p_4), \tag{3.43}$$

with vertices V_1^μ and V_2^ν respectively from Feynman rules given by eqs. (2.20, 2.21). Writing down V_2^ν explicitly:

$$V_2^\nu = \frac{2i}{v}\delta_{IJ}m_{\nu_{f_1}},$$

one may now argue, that since this term is proportional to neutrino mass $V_2^\nu \propto m_{\nu_{f_1}}$ and it was assumed that neutrinos are massless, the amplitude vanishes:

$$i\mathscr{M}_h = 0. \tag{3.44}$$

3.3. Total amplitude

Summarising the results of the previous sections, the total amplitude for muon decay process in SMEFT reads:

$$\begin{aligned}
i\mathscr{M}_{total} = {} & \phi_1 \bar{u}_e(p_3)\gamma^\mu P_L u_\mu(p_1)\bar{u}_{\nu_J}(p_2)\gamma_\mu\gamma^5 v_{\bar{\nu}_I}(p_4) \\
& +\phi_2 \bar{u}_e(p_3)\gamma^\mu P_R u_\mu(p_1)\bar{u}_{\nu_J}(p_2)\gamma_\mu\gamma^5 v_{\bar{\nu}_I}(p_4) \\
& +\phi_3 \bar{u}_e(p_3) P_R u_\mu(p_1)\bar{u}_{\nu_J}(p_2)\not{p}_3\gamma^5 v_{\bar{\nu}_I}(p_4) \\
& +\phi_4 \bar{u}_e(p_3) P_L u_\mu(p_1)\bar{u}_{\nu_J}(p_2)\not{p}_3\gamma^5 v_{\bar{\nu}_I}(p_4) \\
& +(D_{31}+D_1)\bar{u}_{\nu_J}(p_2)\gamma^\mu P_L u_\mu(p_1)\bar{u}_e(p_3)\gamma_\mu P_L v_{\bar{\nu}_I}(p_4) \\
& +D_{21}\bar{u}_{\nu_J}(p_2)\not{p}_3 P_L u_\mu(p_1)\bar{u}_e(p_3) P_L v_{\bar{\nu}_I}(p_4) \\
& -D_{22}\bar{u}_{\nu_J}(p_2) P_R u_\mu(p_1)\bar{u}_e(p_3) P_L v_{\nu_I}(p_4) \\
& -D_{32}\bar{u}_{\nu_J}(p_2) P_R u_\mu(p_1)\bar{u}_e(p_3)\not{p}_2 P_L v_{\nu_I}(p_4) \\
& +F_1 \bar{u}_e(p_3)\gamma^\mu P_L u_\mu(p_1)\bar{u}_{\nu_J}(p_2)\gamma_\mu P_L v_{\bar{\nu}_I}(p_4) \\
& -F_2 \bar{u}_e(p_3)\gamma^\mu P_L u_\mu(p_1)\bar{u}_{\nu_J}(p_2)\gamma_\mu P_R v_{\bar{\nu}_I}(p_4) \\
& +F_3 \bar{u}_e(p_3)\gamma^\mu P_R u_\mu(p_1)\bar{u}_{\nu_J}(p_2)\gamma_\mu P_L v_{\bar{\nu}_I}(p_4) \\
& -F_4 \bar{u}_e(p_3)\gamma^\mu P_R u_\mu(p_1)\bar{u}_{\nu_J}(p_2)\gamma_\mu P_R v_{\bar{\nu}_I}(p_4),
\end{aligned} \tag{3.45}$$

Two additional simplifications can be obtained using Fierz identities:

1. using eq. (A.38):

$$F_1 \bar{u}_e(p_3)\gamma^\mu P_L u_\mu(p_1)\bar{u}_{\nu_J}(p_2)\gamma_\mu P_L v_{\bar{\nu}_I}(p_4) = F_1 \bar{u}_{\nu_J}(p_2)\gamma^\mu P_L u_\mu(p_1)\bar{u}_e(p_3)\gamma_\mu P_L v_{\bar{\nu}_I}(p_4),$$

   which can be grouped with:

$$(D_1 + D_{31} + F_1)\bar{u}_{\nu_J}(p_2)\gamma^\mu P_L u_\mu(p_1)\bar{u}_e(p_3)\gamma_\mu P_L v_{\bar{\nu}_I}(p_4),$$

2. using eq. (A.37):

$$F_3 \bar{u}_e(p_3)\gamma^\mu P_R u_\mu(p_1)\bar{u}_{\nu_J}(p_2)\gamma_\mu P_L v_{\bar{\nu}_I}(p_4) = -2F_3 \bar{u}_{\nu_J}(p_2) P_R u_\mu(p_1)\bar{u}_e(p_3) P_L v_{\nu_I}(p_4),$$

   which can be grouped with:

$$-(D_{22} + 2F_3)\bar{u}_{\nu_J}(p_2) P_R u_\mu(p_1)\bar{u}_e(p_3) P_L v_{\nu_I}(p_4),$$

This simplifications allows to write expression (3.45) in the following form:

$$\begin{aligned}
i\mathscr{M}_{total} = {} & \phi_1 \bar{u}_e(p_3)\gamma^\mu P_L u_\mu(p_1)\bar{u}_{\nu_J}(p_2)\gamma_\mu\gamma^5 v_{\bar{\nu}_I}(p_4)\ ]M_1 \\
& +\phi_2 \bar{u}_e(p_3)\gamma^\mu P_R u_\mu(p_1)\bar{u}_{\nu_J}(p_2)\gamma_\mu\gamma^5 v_{\bar{\nu}_I}(p_4)\ ]M_2 \\
& +\phi_3 \bar{u}_e(p_3) P_R u_\mu(p_1)\bar{u}_{\nu_J}(p_2)\not{p}_3\gamma^5 v_{\bar{\nu}_I}(p_4)\ ]M_3 \\
& +\phi_4 \bar{u}_e(p_3) P_L u_\mu(p_1)\bar{u}_{\nu_J}(p_2)\not{p}_3\gamma^5 v_{\bar{\nu}_I}(p_4)\ ]M_4 \\
& +(D_{31}+D_1+F_1)\bar{u}_{\nu_J}(p_2)\gamma^\mu P_L u_\mu(p_1)\bar{u}_e(p_3)\gamma_\mu P_L v_{\bar{\nu}_I}(p_4)\ ]M_5 \\
& +D_{21}\bar{u}_{\nu_J}(p_2)\not{p}_3 P_L u_\mu(p_1)\bar{u}_e(p_3) P_L v_{\bar{\nu}_I}(p_4)\ ]M_6 \\
& -(D_{22}+2F_3)\bar{u}_{\nu_J}(p_2) P_R u_\mu(p_1)\bar{u}_e(p_3) P_L v_{\nu_I}(p_4)\ ]M_7 \\
& -D_{32}\bar{u}_{\nu_J}(p_2) P_R u_\mu(p_1)\bar{u}_e(p_3)\not{p}_2 P_L v_{\nu_I}(p_4)\ ]M_8 \\
& -F_2 \bar{u}_e(p_3)\gamma^\mu P_L u_\mu(p_1)\bar{u}_{\nu_J}(p_2)\gamma_\mu P_R v_{\bar{\nu}_I}(p_4)\ ]M_9 \\
& -F_4 \bar{u}_e(p_3)\gamma^\mu P_R u_\mu(p_1)\bar{u}_{\nu_J}(p_2)\gamma_\mu P_R v_{\bar{\nu}_I}(p_4)\ ]M_{10}.
\end{aligned} \tag{3.46}$$

All constants are collected in Table 3.3. Symbols M_i in eq. (3.46) are introduced to compactify notation in further calculations.

Table 3.1: Constants used to express the total amplitude.

Z^0	$W^\pm$	4-fermion
$\phi_1 = B_3(C_{12}^{\varphi l_1} + C_{12}^{\varphi l_3})$	$D_1 = D_{SM} = \frac{-i\bar{g}^2 U_{2J}^* U_{1I}}{2M_W^2}$	$F_1 = 2iU_{g_4 I} U_{g_3 J}^* C_{12 g_3 g_4}^{ll}$
$+2B_1 m_\mu C_{12}^{eW}$	$D_{21} = \frac{-i2\sqrt{2}v\bar{g}U_{2J}^* U_{g_2 I} C_{1g_2}^{eW*}}{M_W^2}$	$F_2 = 2iU_{g_3 J} U_{g_4 I}^* C_{12 g_4 g_3}^{ll}$
$+2B_2 m_\mu C_{12}^{eB}$	$D_{22} = \frac{-i\sqrt{2}vm_\mu\bar{g}U_{2J}^* U_{g_2 I} C_{1g_2}^{eW*}}{M_W^2}$	$F_3 = iU_{g_4 I} U_{g_3 J}^* C_{g_3 g_4 12}^{le}$
$\phi_2 = B_3 C_{12}^{\varphi l_3} + 2B_1 m_\mu C_{21}^{eW*}$	$D_{31} = \frac{i\sqrt{2}vm_\mu\bar{g}U_{g_1 J}^* U_{1I} C_{g_1 2}^{eW}}{M_W^2}$	$F_4 = iU_{g_3 J} U_{g_4 I}^* C_{g_4 g_3 12}^{l3}$
$+2B_2 m_\mu C_{21}^{eB*}$	$D_{32} = \frac{i2\sqrt{2}v\bar{g}U_{g_1 J}^* U_{1I} C_{g_1 2}^{eW}}{M_W^2}$	
$\phi_3 = -4(B_1 C_{12}^{eW} + B_2 C_{12}^{eB})$		
$\phi_4 = -4(B_1 C_{21}^{eW*} + B_2 C_{21}^{eB*})$		
$B_1 = \frac{i\sqrt{2}\bar{g}v\delta_{IJ}}{4M_Z^2}$		
$B_2 = \frac{i\sqrt{2}\bar{g}'v\delta_{IJ}}{4M_Z^2}$		
$B_3 = \frac{-i(\bar{g}^2 + \bar{g}'^2)v^2\delta_{IJ}}{4M_Z^2}$		

3.4. Amplitude squared

In the next step, one needs to calculate the amplitude squared which reads:

$$|\mathscr{M}|^2 = \mathscr{M}_{total}^\dagger \mathscr{M}_{total}. \tag{3.47}$$

At first, one should note that Standard Model part of expression (3.46) is included in M_5 part and reads:

$$M_{SM} = D_1 \bar{u}(p_2)\gamma^\mu P_L u(p_1)\bar{u}(p_3)\gamma_\mu P_L v(p_4), \tag{3.48}$$

and M_5 can be separated to:

$$\begin{aligned} M_5 = D_1 \bar{u}(p_2)\gamma^\mu P_L u(p_1)\bar{u}(p_3)\gamma_\mu P_L v(p_4)\ ]M_{SM} \\ +(D_{31} + F_1)\bar{u}(p_2)\gamma^\mu P_L u(p_1)\bar{u}(p_3)\gamma_\mu P_L v(p_4)\ ]M_{5_1}, \end{aligned} \tag{3.49}$$

total amplitude expression (3.46) can be now rewritten in the following form:

$$i\mathscr{M}_{total} = M_{SM} + M_{SMEFT}. \tag{3.50}$$

M_{SMEFT} contains all expressions except (3.48).
Putting eq. (3.50) into eq. (3.47) one gets:

$$|\mathscr{M}|^2 = M_{SM}^\dagger M_{SM} + M_{SMEFT}^\dagger M_{SM} + M_{SM}^\dagger M_{SMEFT} + M_{SMEFT}^\dagger M_{SMEFT}. \tag{3.51}$$

Since every M_{SMEFT} term is proportional Wilson coefficients C, $M_{SMEFT}^\dagger M_{SMEFT} \propto C^2 = 0$ and the whole expression for squared amplitude simplifies to:

$$\begin{aligned} |\mathscr{M}|^2 = M_{SM}^\dagger M_{SM} + M_{SMEFT}^\dagger M_{SM} + M_{SM}^\dagger M_{SMEFT} = \\ M_{SM}^\dagger M_{SM} + 2\Re(M_{SM}^\dagger M_{SMEFT}). \end{aligned} \tag{3.52}$$

It is also important to take care of spins. This means that one has to:

1. average the squared amplitude over initial spin s_1,

2. sum over the final spins s_2, s_3, s_4,

as a consequence, one has to calculate the following expression:

$$|\bar{\mathscr{M}}|^2 = \frac{1}{2}\sum_{s_{1,2,3,4}} |\mathscr{M}|^2 = \frac{1}{2}\sum_{s_{1,2,3,4}} \left(M_{SM}^\dagger M_{SM} + 2\Re(M_{SM}^\dagger M_{SMEFT})\right). \tag{3.53}$$

At first, it is convenient to calculate Standard Model part of expression (3.53). Noting that:

$$M_{SM}^\dagger = D_1^\dagger \bar{\nu}(p_4) P_R \gamma_\mu u(p_3) \bar{u}(p_1) P_R \gamma^\mu u(p_2), \tag{3.54}$$

SM part of eq. (3.53) then reads:

$$\frac{1}{2}\sum_{s_{1,2,3,4}} M_{SM}^\dagger M_{SM} =$$
$$\frac{1}{2}\sum_{s_{1,2,3,4}} D_1^\dagger D_1 \bar{v}(p_4) P_R \gamma_\nu u(p_3) \bar{u}(p_1) P_R \gamma^\nu u(p_2) \bar{u}(p_2) \gamma^\mu P_L u(p_1) \bar{u}(p_3) \gamma_\mu P_L v(p_4), \tag{3.55}$$

it is convenient to introduce indices:

$$\frac{1}{2}\sum_{s_{1,2,3,4}} M_{SM}^\dagger M_{SM} =$$
$$\frac{1}{2}\sum_{s_{1,2,3,4}} D_1^\dagger D_1 \bar{v}(p_4)_m (P_R\gamma_\nu)_{mn} u(p_3)_n \bar{u}(p_1)_o (P_R\gamma^\nu)_{op} u(p_2)_p \bar{u}(p_2)_i (\gamma^\mu P_L)_{ij} u(p_1)_j \bar{u}(p_3)_k (\gamma_\mu P_L)_{kl} v(p_4)_l,$$

since one does not have to worry about commutations, above expression (3.4) can be rewritten in the following form:

$$\frac{1}{2}\sum_{s_{1,2,3,4}} M_{SM}^\dagger M_{SM} =$$
$$\frac{1}{2} D_1 D_1^\dagger \sum_{s_1 s_2} (u(p_1)_j \bar{u}(p_1)_o (P_R\gamma^\nu)_{op} u(p_2)_p \bar{u}(p_2)_i (\gamma^\mu P_L)_{ij}) \sum_{s_3 s_4} (v(p_4)_l \bar{v}(p_4)_m (P_R\gamma_\nu)_{mn} u(p_3)_n \bar{u}(p_3)_k (\gamma_\mu P_L)_{kl}),$$

one may argue that since in both sums there are multiplied matrices and spinors with the same indices in the beginning and end, these are just traces:

$$\frac{1}{2}\sum_{s_{1,2,3,4}} M_{SM}^\dagger M_{SM} =$$
$$\frac{1}{2} D_1^\dagger D_1 \sum_{s_1 s_2} (u(p_1)_j \bar{u}(p_1)_o (P_R\gamma^\nu)_{op} u(p_2)_p \bar{u}(p_2)_i (\gamma^\mu P_L)_{ij}) \sum_{s_3 s_4} (v(p_4)_l \bar{v}(p_4)_m (P_R\gamma_\nu)_{mn} u(p_3)_n \bar{u}(p_3)_k (\gamma_\mu P_L)_{kl}) =$$
$$\frac{1}{2} D_1^\dagger D_1 \sum_{s_1 s_2} \mathrm{tr}\,(u(p_1)\bar{u}(p_1) P_R \gamma^\nu u(p_2) \bar{u}(p_2) \gamma^\mu P_L) \sum_{s_3 s_4} \mathrm{tr}\,(v(p_4)\bar{v}(p_4) P_R \gamma_\nu u(p_3) \bar{u}(p_3) \gamma_\mu P_L),$$

using spin sum formulas given by eqs. (3.6, 3.7, 3.8):

$$\frac{1}{2} D_1^\dagger D_1 \mathrm{tr}\,((\not{p_1} + m_\mu) P_R \gamma^\nu \not{p_2} \gamma^\mu P_L)\, \mathrm{tr}\,(\not{p_4} P_R \gamma_\nu \not{p_3} \gamma_\mu P_L),$$

one can consider both traces separately.
For the first one, using commutation relation (A.11):

$$\mathrm{tr}\,((\not{p_1} + m_\mu) P_R \gamma^\nu \not{p_2} \gamma^\mu P_L) =$$
$$\mathrm{tr}\,(\not{p_1} \gamma^\nu \not{p_2} \gamma^\mu P_L) + \mathrm{tr}\,(m_\mu \gamma^\nu \not{p_2} \gamma^\mu P_L),$$

mass term has odd number of γ matrices, so from eq. (A.24) it vanishes. Remaining term reads:

$$\frac{1}{2} p_{1\alpha} p_{2\beta} \mathrm{tr}(\gamma^\alpha \gamma^\mu \gamma^\beta \gamma^\nu - \gamma^\alpha \gamma^\mu \gamma^\beta \gamma^\nu \gamma^5),$$

this gives from eqs. (A.26, A.29):

$$2 p_{1\alpha} p_{2\beta} (\eta^{\alpha\mu}\eta^{\beta\nu} - \eta^{\alpha\beta}\eta^{\mu\nu} + \eta^{\alpha\nu}\eta^{\beta\nu} + i\epsilon^{\alpha\mu\beta\nu}),$$

second trace, similarly from eqs. (A.26, A.29) reads:

$$\mathrm{tr}\left(\not{p_3}\gamma_\mu P_L \not{p_4} P_R\gamma_\nu\right) = p_3^\sigma p_4^\rho \mathrm{tr}\left(\gamma_\sigma\gamma_\mu\gamma_\rho\gamma_\nu P_L\right) = \frac{1}{2}p_3^\sigma p_4^\rho \mathrm{tr}\left(\gamma_\sigma\gamma_\mu\gamma_\rho\gamma_\nu - \gamma_\sigma\gamma_\mu\gamma_\rho\gamma_\nu\gamma^5\right) =$$
$$2p_3^\sigma p_4^\rho(\eta_{\sigma\mu}\eta_{\rho\nu} - \eta_{\sigma\rho}\eta_{\mu\nu} + \eta_{\sigma\nu}\eta_{\rho\nu} + i\epsilon_{\sigma\mu\rho\nu}),$$

whole expression reads:

$$\frac{1}{2}D_1^\dagger D_1 \sum_{s_1 s_2} \mathrm{tr}\left(u(p_2)\bar{u}(p_2)\gamma^\mu P_L u(p_1)\bar{u}(p_1)P_R\gamma^\nu\right) \sum_{s_3 s_4} \mathrm{tr}\left(u(p_3)\bar{u}(p_3)\gamma_\mu P_L v(p_4)\bar{v}(p_4) P_R\gamma_\nu\right) = \tag{3.56}$$
$$2D_1^\dagger D_1 p_{1\alpha}p_{2\beta}(\eta^{\alpha\mu}\eta^{\beta\nu} - \eta^{\alpha\beta}\eta^{\mu\nu} + \eta^{\alpha\nu}\eta^{\beta\nu} + i\epsilon^{\alpha\mu\beta\nu})p_3^\sigma p_4^\rho(\eta_{\sigma\mu}\eta_{\rho\nu} - \eta_{\sigma\rho}\eta_{\mu\nu} + \eta_{\sigma\nu}\eta_{\rho\nu} + i\epsilon_{\sigma\mu\rho\nu}) =$$
$$2D_1 D_1^\dagger(p_1^\mu p_2^\nu - (p_1\cdot p_2)\eta^{\mu\nu} + p_1^\nu p_2^\mu + i\epsilon^{\alpha\mu\beta\nu}p_{1\alpha}p_{2\beta})(p_{3\mu}p_{4\nu} - (p_3\cdot p_4)\eta_{\mu\nu} + p_{3\nu}p_{4\mu} + i\epsilon_{\sigma\mu\rho\nu}p_3^\sigma p_4^\rho),$$

since first three terms in both brackets are even and last one containing ϵ is odd, only the following terms will survive:

$$2D_1^\dagger D_1((p_1^\mu p_2^\nu - (p_1\cdot p_2)\eta^{\mu\nu} + p_1^\nu p_2^\mu)(p_{3\mu}p_{4\nu} - (p_3\cdot p_4)\eta_{\mu\nu} + p_{3\nu}p_{4\mu}) - \epsilon^{\alpha\mu\beta\nu}\epsilon_{\sigma\mu\rho\nu}p_{1\alpha}p_{2\beta}p_3^\sigma p_4^\rho),$$

after multiplication, simplification and use of eq. (A.19)to above expression, one obtains the final result:

$$\frac{1}{2}\sum_{s_{1,2,3,4}} \mathscr{M}_{SM}\mathscr{M}_{SM}^\dagger = 8D_1 D_1^\dagger 8(p_1\cdot p_4)(p_2\cdot p_3). \tag{3.57}$$

One may easily notice that whole expression M_5 (3.49) has the same form as M_{SM} (3.48), so the result will differ only in constants and therefore reads:

$$\frac{1}{2}\sum_{s_{1,2,3,4}}(M_{SM}^\dagger M_{SM} + 2\Re(M_{SM}^\dagger M_{51})) = (8D_1^\dagger D_1 + 16\Re(D_1^\dagger D_{31} + 16\Re(D_1^\dagger F_1))(p_1\cdot p_4)(p_2\cdot p_3). \tag{3.58}$$

The same procedure as above will be performed to all other terms in expression (3.46) from first term, M_1 onwards.
Squared amplitude, summed over spins for M_1:

$$\sum_{s_{1,2,3,4}}\Re(M_{SM}^\dagger M_1) = \Re(D_1^\dagger\phi_1)\sum_{s_{1,2,3,4}}(\bar{v}(p_4)P_R\gamma_\mu u_e(p_3)\bar{u}(p_1)P_R\gamma^\mu u(p_2)\bar{u}(p_3)\gamma^\mu P_L u(p_1)\bar{u}(p_2)\gamma_\mu\gamma^5 v(p_4)) =$$
$$\Re(D_1^\dagger\phi_1)\sum_{s_{1,2,3,4}}(\bar{v}(p_4)_a(P_R\gamma_\mu)_{ab}u(p_3)_b\bar{u}(p_1)_c(P_R\gamma^\mu)_{cd}u(p_2)_d\bar{u}(p_3)_e(\gamma^\nu P_L)_{ef}u(p_1)_f\bar{u}(p_2)_g(\gamma_\nu\gamma^5)_{gh}v(p_4)_h) =$$
$$\Re(D_1^\dagger\phi_1)(v(p_4)_h\bar{v}(p_4)_a(P_R\gamma_\mu)_{ab}u(p_3)_b\bar{u}(p_3)_e(\gamma^\nu P_L)_{ef}u(p_1)_f\bar{u}(p_1)_c(P_R\gamma^\mu)_{cd}u(p_2)_d\bar{u}(p_2)_g(\gamma_\nu\gamma^5)_{gh}) =$$
$$\Re(D_1^\dagger\phi_1)\mathrm{tr}(\not{p_4}P_R\gamma_\mu\not{p_3}\gamma^\nu P_L(\not{p_1} + m_\mu)P_R\gamma^\mu\not{p_2}\gamma_\nu\gamma^5) =$$
$$\Re(D_1^\dagger\phi_1)(\mathrm{tr}(\not{p_4}P_R\gamma_\mu\not{p_3}\gamma^\nu P_L\not{p_1}P_R\gamma^\mu\not{p_2}\gamma_\nu\gamma^5) + \mathrm{tr}(m_\mu\not{p_4}P_R\gamma_\mu\not{p_3}\gamma^\nu P_L P_R\gamma^\mu\not{p_2}\gamma_\nu\gamma^5)),$$

mass term contains odd number of γ matrices, so from eq. (A.24) vanishes. Remaining part can be simplified using eq. (A.11) to:

$$-\Re(D_1^\dagger\phi_1)p_{4\alpha}p_{3\beta}p_{1\sigma}p_{2\rho}\mathrm{tr}(\gamma^\alpha\gamma_\mu\gamma^\beta\gamma^\nu\gamma^\sigma\gamma^\mu\gamma^\rho\gamma_\nu P_L),$$

from eqs. (A.18, A.17) one obtains:

$$8\Re(D_1^\dagger\phi_1)p_{4\alpha}p_{3\beta}p_{1\sigma}p_{2\rho}\eta^{\beta\rho}\mathrm{tr}(\gamma^\alpha\gamma^\sigma P_L) = 4\Re(D_1^\dagger\phi_1)p_{4\alpha}p_{3\beta}p_{1\sigma}p_{2\rho}\eta^{\beta\rho}(\mathrm{tr}((\gamma^\alpha\gamma^\sigma) - \mathrm{tr}(\gamma^\alpha\gamma^\sigma\gamma^5)),$$

at last, using (A.25, A.28) one arrives to the final result:

$$\sum_{s_{1,2,3,4}}\Re(M_{SM}^\dagger M_1) = 16\Re(D_1^\dagger\phi_1)(p_1\cdot p_4)(p_2\cdot p_3). \tag{3.59}$$

Subsequent term, which one needs to calculate has the form:

$$\sum_{s_{1,2,3,4}}\Re(M_{SM}^\dagger M_2) = \Re(D_1^\dagger\phi_2)\sum_{s_{1,2,3,4}}\bar{v}(p_4)P_R\gamma_\mu u(p_3)\bar{u}(p_1)P_R\gamma^\mu u(p_2)\bar{u}(p_3)\gamma^\nu P_R u(p_1)\bar{u}(p_2)\gamma_\nu\gamma^5 v(p_4) =$$
$$\Re(D_1^\dagger\phi_2)\mathrm{tr}(\not{p_4}P_R\gamma_\mu\not{p_3}\gamma^\nu P_R(\not{p_1} + m_\mu)P_R\gamma^\mu\not{p_2}\gamma_\nu\gamma^5),$$

one can now use commutation relation given by eq. (A.11) and rewrite above expression:

$$\Re(D_1^\dagger\phi_2)\mathrm{tr}(\not{p_4}\gamma_\mu\not{p_3}\gamma^\nu P_L P_R(\not{p_1}+m_\mu)P_R\gamma^\mu\not{p_2}\gamma_\nu\gamma^5),$$

from eq. (A.12) above term vanishes, so finally:

$$\sum_{s_{1,2,3,4}}\Re(M_{SM}^\dagger M_2)=0. \tag{3.60}$$

Next expression reads:

$$\sum_{s_{1,2,3,4}}\Re(M_{SM}^\dagger M_3)=\Re(D_1^\dagger\phi_3)\sum_{s_{1,2,3,4}}\bar{v}(p_4)P_R\gamma_\mu u(p_3)\bar{u}(p_1)P_R\gamma^\mu u(p_2)\bar{u}(p_3)P_R u(p_1)\bar{u}(p_2)\not{p_3}\gamma^5 v(p_4)=$$
$$\Re(D_1^\dagger\phi_3)\mathrm{tr}(\not{p_4}P_R\gamma_\mu\not{p_3}P_R(\not{p_1}+m_\mu)P_R\gamma^\mu\not{p_2}\not{p_3}\gamma^5)=$$
$$\Re(D_1^\dagger\phi_3)(\mathrm{tr}(\not{p_4}P_R\gamma_\mu\not{p_3}P_R\not{p_1}P_R\gamma^\mu\not{p_2}\not{p_3}\gamma^5)+\mathrm{tr}(\not{p_4}P_R\gamma_\mu\not{p_3}P_R m_\mu P_R\gamma^\mu\not{p_2}\not{p_3}\gamma^5)),$$

using commutation relation (A.11) one can rewrite the above equation:

$$\sum_{s_{1,2,3,4}}\Re(M_{SM}^\dagger M_3)=\Re(D_1^\dagger\phi_3)(\mathrm{tr}(\not{p_4}P_R\gamma_\mu\not{p_3}\not{p_1}P_L P_R\gamma^\mu\not{p_2}\not{p_3}\gamma^5)-\mathrm{tr}(\not{p_4}\gamma_\mu\not{p_3}m_\mu\gamma^\mu\not{p_2}\not{p_3}P_L)),$$

one can now note, that from eq. (A.12) only mass term survives:

$$-\Re(D_1^\dagger\phi_3)m_\mu\mathrm{tr}(\not{p_4}\gamma_\mu\not{p_3}m_\mu\gamma^\mu\not{p_2}\not{p_3}P_L)=-\Re(D_1^\dagger\phi_3)m_\mu p_{4\alpha}p_{3\beta}p_{4\sigma}p_{3\rho}\mathrm{tr}(\gamma^\alpha\gamma^\mu\gamma^\beta\gamma_\mu\gamma^\sigma\gamma^\rho P_L),$$

using eq. (A.16) one gets:

$$\sum_{s_{1,2,3,4}}\Re(M_{SM}^\dagger M_3)=\Re(D_1^\dagger\phi_3)m_\mu(\mathrm{tr}(\gamma^\alpha\gamma^\beta\gamma^\sigma\gamma^\rho)-\mathrm{tr}(\gamma^\alpha\gamma^\beta\gamma^\sigma\gamma^\rho\gamma^5)),$$

and from eqs. (A.26, A.29):

$$\sum_{s_{1,2,3,4}}\Re(M_{SM}^\dagger M_3)=4\Re(D_1^\dagger\phi_3)m_\mu p_{4\alpha}p_{3\beta}p_{2\sigma}p_{3\rho}(\eta^{\alpha\beta}\eta^{\sigma\rho}-\eta^{\alpha\sigma}\eta^{\beta\rho}+\eta^{\alpha\rho}\eta^{\beta\sigma}-i\epsilon^{\alpha\beta\sigma\rho})=$$
$$4\Re(D_1^\dagger\phi_3)m_\mu(2(p_2\cdot p_3)(p_4\cdot p_3)-(p_2\cdot p_4)(p_3\cdot p_3)-i\epsilon^{\alpha\beta\sigma\rho}p_{4\alpha}p_{3\beta}p_{2\sigma}p_{3\rho}),$$

ϵ-term is zero, because is antisymmetric, and $p_{3\beta}p_{3\rho}$ is symmetric in β and ρ. Moreover, since $m_e=0$, $p_3\cdot p_3=0$.
All in all the final result reads:

$$\sum_{s_{1,2,3,4}}\Re(M_{SM}^\dagger M_3)=8\Re(D_1^\dagger\phi_3)m_\mu(p_2\cdot p_3)(p_4\cdot p_3). \tag{3.61}$$

Following term needs now to be calculated:

$$\sum_{s_{1,2,3,4}}\Re(M_{SM}^\dagger M_4)=\Re(D_1^\dagger\phi_4)\sum_{s_{1,2,3,4}}\bar{v}(p_4)P_R\gamma_\mu u(p_3)\bar{u}(p_1)P_R\gamma^\mu u(p_2)\bar{u}(p_3)P_L u(p_1)\bar{u}(p_2)\not{p_3}\gamma^5 v(p_4)=$$
$$\Re(D_1^\dagger\phi_4)\mathrm{tr}(\not{p_4}P_R\gamma_\mu\not{p_3}P_L(\not{p_1}+m_\mu)P_R\gamma^\mu\not{p_2}\not{p_3}\gamma^5),$$

using commutation relation (A.11):

$$\Re(D_1^\dagger\phi_4)\mathrm{tr}(\not{p_4}\gamma_\mu\not{p_3}P_R P_L(\not{p_1}+m_\mu)P_R\gamma^\mu\not{p_2}\not{p_3}\gamma^5),$$

from eq. (A.12) the whole expression vanishes, so finally:

$$\sum_{s_{1,2,3,4}}\Re(M_{SM}^\dagger M_4)=0. \tag{3.62}$$

Since $\mathscr{M}_5$ term has been already calculated, eq. (3.58), next term then reads:

$$\sum_{s_{1,2,3,4}} \Re(M_{SM}^\dagger M_6) = \Re(D_1^\dagger D_{21}) \sum_{s_{1,2,3,4}} \bar{v}(p_4) P_R \gamma_\mu u(p_3) \bar{u}(p_1) P_R \gamma^\mu u(p_2) \bar{u}(p_2) \not{p_3} P_L u(p_1) \bar{u}(p_3) P_L v(p_4) =$$
$$\Re(D_1^\dagger D_{21}) \mathrm{tr}(\not{p_4} P_R \gamma_\mu \not{p_3} P_L) \mathrm{tr}((\not{p_1} + m_\mu) P_R \gamma^\mu \not{p_2} \not{p_3} P_L),$$

once again, applying commutation relation (A.11):

$$\frac{1}{2} \mathrm{tr}(\not{p_4} \gamma_\mu \not{p_3} P_R P_L) \mathrm{tr}((\not{p_1} + m_\mu) P_R \gamma^\mu \not{p_2} \not{p_3} P_L),$$

using eq. (A.12) the whole expression vanishes and one obtains the final result:

$$\sum_{s_{1,2,3,4}} \Re(M_{SM}^\dagger M_6) = 0. \tag{3.63}$$

Subsequent expression has form:

$$\sum_{s_{1,2,3,4}} \Re(M_{SM}^\dagger M_7) = -\Re(D_1^\dagger D_{22}) \sum_{s_{1,2,3,4}} \bar{v}(p_4) P_R \gamma_\mu u(p_3) \bar{u}(p_1) P_R \gamma^\mu u(p_2) \bar{u}(p_2) P_R u(p_1) \bar{u}(p_3) P_L v(p_4) =$$
$$-\Re(D_1^\dagger D_{22}) \mathrm{tr}(\not{p_4} P_R \gamma_\mu \not{p_3} P_L) \mathrm{tr}((\not{p_1} + m_\mu) P_R \gamma^\mu \not{p_2} P_R),$$

again, from commutation relation (A.11) one can obtain:

$$\sum_{s_{1,2,3,4}} \Re(M_{SM}^\dagger M_7) = -\Re(D_1^\dagger D_{22}) \mathrm{tr}(\not{p_4} \gamma_\mu \not{p_3} P_R P_L) \mathrm{tr}((\not{p_1} + m_\mu) \gamma^\mu \not{p_2} P_R),$$

using eq. (A.12) the whole term vanishes, so the result reads:

$$\sum_{s_{1,2,3,4}} \Re(M_{SM}^\dagger M_7) = 0. \tag{3.64}$$

Next term reads:

$$\sum_{s_{1,2,3,4}} \Re(M_{SM}^\dagger M_8) = -\Re(D_1^\dagger D_{32}) \sum_{s_{1,2,3,4}} \bar{v}(p_4) P_R \gamma_\mu u(p_3) \bar{u}(p_1) P_R \gamma^\mu u(p_2) \bar{u}(p_2) P_R u(p_1) \bar{u}(p_3) \not{p_2} P_L v(p_4) =$$
$$-\Re(D_1^\dagger D_{32}) \mathrm{tr}(\not{p_4} P_R \gamma_\mu \not{p_3} \not{p_2} P_L) \mathrm{tr}((\not{p_1} + m_\mu) P_R \gamma^\mu \not{p_2} P_R),$$

applying commutation relation (A.11) and multiplying:

$$\sum_{s_{1,2,3,4}} \Re(M_{SM}^\dagger M_8) = -\Re(D_1^\dagger D_{32}) \mathrm{tr}(\not{p_4} \gamma_\mu \not{p_3} \not{p_2} P_L)(\mathrm{tr}(\not{p_1} \gamma^\mu \not{p_2} P_R) + \mathrm{tr}(m_\mu \gamma^\mu \not{p_2} P_R)),$$

due to eq. (A.24) only mass terms survives and then:

$$\sum_{s_{1,2,3,4}} \Re(M_{SM}^\dagger M_8) = -\Re(D_1^\dagger D_{32}) m_\mu p_4^\alpha p_3^\beta p_2^\sigma p_{2\rho} \mathrm{tr}(\gamma_\alpha \gamma_\mu \gamma_\beta \gamma_\sigma P_L) \mathrm{tr}(\gamma^\mu \gamma^\rho P_R),$$

writing explicitly P_L, P_R and simplifying:

$$\sum_{s_{1,2,3,4}} \Re(M_{SM}^\dagger M_8) = -\Re(D_1^\dagger D_{32}) m_\mu p_4^\alpha p_3^\beta p_2^\sigma p_{2\rho} (\mathrm{tr}(\gamma_\alpha \gamma_\mu \gamma_\beta \gamma_\sigma) - \mathrm{tr}(\gamma_\alpha \gamma_\mu \gamma_\beta \gamma_\sigma \gamma^5)) \eta^{\rho\mu},$$

using eqs. (A.26, A.29) one obtains:

$$\sum_{s_{1,2,3,4}} \Re(M_{SM}^\dagger M_8) = -4\Re(D_1^\dagger D_{32}) m_\mu p_4^\alpha p_3^\beta p_2^\sigma p_2^\mu (\eta^{\alpha\mu} \eta^{\beta\sigma} - \eta^{\alpha\beta} \eta^{\mu\sigma} + \eta^{\alpha\sigma} \eta^{\mu\beta} - i\epsilon^{\alpha\mu\beta\sigma}) =$$
$$-8\Re(D_1^\dagger D_{32}) m_\mu (2(p_2 \cdot p_4)(p_3 \cdot p_2) - (p_4 \cdot p_3)(p_2 \cdot p_2) - i\epsilon^{\alpha\mu\beta\sigma} p_4^\alpha p_3^\beta p_2^\sigma p_2^\mu),$$

ϵ-term vanishes because is antisymmetric and $p_2^\sigma p_2^\mu$ is symmetric in indices, moreover since $m_\nu = 0$, $p_2 \cdot p_2 = 0$. Finally, one arrives to the final formula:

$$\sum_{s_{1,2,3,4}} \Re(M_{SM}^\dagger M_8) = -8\Re(D_1^\dagger D_{32}) m_\mu (p_2 \cdot p_4)(p_3 \cdot p_2). \tag{3.65}$$

There are two terms left, the first one has following form:

$$\sum_{s_{1,2,3,4}} \Re(M_{SM}^\dagger M_9) = -\Re(D_1^\dagger F_2) \sum_{s_{1,2,3,4}} \bar{v}(p_4) P_R \gamma_\mu u(p_3) \bar{u}(p_1) P_R \gamma^\mu u(p_2) \bar{u}(p_3) \gamma^\nu P_L u(p_1) \bar{u}(p_2) \gamma_\nu P_R v(p_4) =$$

$$-\Re(D_1^\dagger F_2) \mathrm{tr}(\not{p}_4 P_R \gamma_\mu \not{p}_3 \gamma^\nu P_L (\not{p}_1 + m_\mu) P_R \gamma^\mu \not{p}_2 \gamma_\nu P_R),$$

after using commutation relation (A.11) one gets:

$$-\Re(D_1^\dagger F_2) \mathrm{tr}(\not{p}_4 P_R \gamma_\mu \not{p}_3 \gamma^\nu P_L (\not{p}_1 + m_\mu) \gamma^\mu \not{p}_2 \gamma_\nu P_L P_R),$$

which vanishes, due to eq. (A.12), so:

$$\sum_{s_{1,2,3,4}} \Re(M_{SM}^\dagger M_9) = 0. \tag{3.66}$$

Last expression that needs to be calculated reads:

$$\sum_{s_{1,2,3,4}} \Re(M_{SM}^\dagger M_{10}) = -\Re(D_1^\dagger F_2) \sum_{s_{1,2,3,4}} \bar{v}(p_4) P_R \gamma_\mu u(p_3) \bar{u}(p_1) P_R \gamma^\mu u(p_2) \bar{u}(p_3) \gamma^\nu P_R u(p_1) \bar{u}(p_2) \gamma_\nu P_R v(p_4) =$$

$$-\Re(D_1^\dagger F_2) \mathrm{tr}(\not{p}_4 P_R \gamma_\mu \not{p}_3 \gamma^\nu P_L (\not{p}_1 + m_\mu) P_R \gamma^\mu \not{p}_2 \gamma_\nu P_R),$$

after application of commutation relation (A.11):

$$\sum_{s_{1,2,3,4}} \Re(M_{SM}^\dagger M_{10}) \mathrm{tr}(\not{p}_4 \gamma_\mu \not{p}_3 \gamma^\nu P_R P_L (\not{p}_1 + m_\mu) \gamma^\mu \not{p}_2 \gamma_\nu P_L P_R),$$

which vanishes from eq. (A.12) and:

$$\sum_{s_{1,2,3,4}} \Re(M_{SM}^\dagger M_{10}) = 0. \tag{3.67}$$

One can now write down the final formula for squared amplitude, averaged over initial and summed over final spins given by remaining terms (3.58, 3.59, 3.61, 3.65):

$$\begin{aligned} |\bar{\mathscr{M}}|^2 = &\Big(8(D_1^\dagger D_1) + 16\Re(D_1^\dagger D_{31}) + 16\Re(D_1^\dagger F_1) + 16\Re(D_1^\dagger \phi_1)\Big)(p_1 \cdot p_4)(p_3 \cdot p_2) \\ &+ 8m_\mu \Re(D_1^\dagger \phi_3)(p_4 \cdot p_3)(p_2 \cdot p_3) - 8m_\mu \Re(D_1^\dagger D_{32})(p_2 \cdot p_4)(p_3 \cdot p_2). \end{aligned} \tag{3.68}$$

3.5. Decay rate

Decay rate formula (A.35) in case of muon decay simplifies to:

$$d\Gamma_\mu = \frac{|\bar{\mathscr{M}}|^2}{2m_\mu} \frac{d^3 p_2}{(2\pi)^3 2E_2} \frac{d^3 p_3}{(2\pi)^3 2E_3} \frac{d^3 p_4}{(2\pi)^3 2E_4} (2\pi)^4 \delta^4 (p_1 - p_2 - p_3 - p_4), \tag{3.69}$$

where $|\bar{\mathscr{M}}|$ refers to eq. (3.68).

3.5.1. Kinematics

Before calculating phase-space integrals, one has to establish kinematics. The easiest case is to work in the rest frame of heavy particle, in this case-muon.
4-momenta and their squares then read:

$$p_1 = (m_\mu, \bar{0}) \to p_1^2 = m_\mu^2, \tag{3.70}$$

$$p_2 = (E_2, \bar{p_2}) \to p_2^2 = E_2^2 - |\bar{p_2}|^2 = m_2^2 = 0, \tag{3.71}$$

$$p_3 = (E_3, \bar{p_3}) \to p_3^2 = E_3^2 - |\bar{p_3}|^2 = m_3^2 = 0, \tag{3.72}$$

$$p_4 = (E_4, \bar{p_4}) \to p_4^2 = E_4^2 - |\bar{p_4}|^2 = m_4^2 = 0. \tag{3.73}$$

3.5.2. Calculations

It is appropriate to introduce some symbols and calculate integrals separately:

$$\begin{aligned} K_1 = \Big(8(D_1^\dagger D_1) + 16\Re(D_1^\dagger D_{31} + D_1^\dagger F_1 + D_1^\dagger F_1)\Big)\,, \\ K_2 = 8m_\mu \Re(D_1^\dagger \phi_3), \\ K_3 = -8m_\mu \Re(D_1^\dagger D_{32}), \end{aligned} \tag{3.74}$$

then:

$$|\bar{\mathscr{M}}|^2 = K_1(p_1\cdot p_4)(p_3\cdot p_2) + K_2(p_4\cdot p_3)(p_2\cdot p_3) + K_3(p_2\cdot p_4)(p_3\cdot p_2) = |\bar{\mathscr{M}}_1|^2 + |\bar{\mathscr{M}}_2|^2 + |\bar{\mathscr{M}}_3|^2.$$

First integral reads:

$$\begin{aligned} d\Gamma_1 = \frac{|\bar{\mathscr{M}}_1|^2}{2m_\mu}\frac{d^3p_2}{(2\pi)^3 2E_2}\frac{d^3p_3}{(2\pi)^3 2E_3}\frac{d^3p_4}{(2\pi)^3 2E_4}(2\pi)^4\delta^4(p_1-p_2-p_3-p_4) = \\ K_1\frac{(p_1\cdot p_4)(p_3\cdot p_2)}{2m_\mu}\frac{d^3p_2}{(2\pi)^3 2E_2}\frac{d^3p_3}{(2\pi)^3 2E_3}\frac{d^3p_4}{(2\pi)^3 2E_4}(2\pi)^4\delta^4(p_1-p_2-p_3-p_4), \end{aligned} \tag{3.75}$$

noting eqs. (3.70-3.73):

$$\begin{aligned} (p_1\cdot p_4) = m_\mu E_4, \\ (p_3+p_2)^2 = p_3^2 + 2(p_3\cdot p_2) + p_2^2 = 2(p_3\cdot p_2), \\ (p_1-p_4)^2 = p_1^2 - 2(p_1\cdot p_4) + p_4^2 = p_1^2 - 2(p_1\cdot p_4) = m_\mu^2 - 2m\mu E_4, \end{aligned}$$

using momentum conservation $p_1 - p_4 = p_2 + p_3$:

$$\begin{aligned} (p_3+p_2)^2 = (p_1-p_4)^2, \\ 2(p_3\cdot p_2) = m_\mu^2 - 2m\mu E_4, \end{aligned}$$

then applying the above results, eq. (3.75) can be rewritten as below:

$$d\Gamma_1 = K_1' E_4(m_\mu - 2E_4)\frac{d^3p_2}{E_2}\frac{d^3p_3}{E_3}\frac{d^3p_4}{E_4}\delta^4(p_1-p_2-p_3-p_4),$$

where

$$K_1' = \frac{m_\mu K_1}{2^{10}\pi^5}.$$

Separating 4-delta function into momentum and energy components:

$$d\Gamma_1 = K_1' E_4(m_\mu - 2E_4)\frac{d^3p_2}{E_2}\frac{d^3p_3}{E_3}\frac{d^3p_4}{E_4}\delta^3(\bar{p_2}+\bar{p_3}+\bar{p_4})\delta(E_1-E_2-E_3-E_4),$$

integrating over d^3p_2 and noting from eq. (3.71) that $E_2 = |\bar{p_2}|$ one arrives to:

$$d\Gamma_1 = K_1' E_4(m_\mu - 2E_4)\frac{d^3p_3 d^3p_4}{|\bar{p_3}+\bar{p_4}|E_3E_4}\delta(m_\mu - |\bar{p_3}+\bar{p_4}| - E_3 - E_4),$$

it is convenient to assume that $\bar{p_4}$ is fixed, then switching p_3 to polar coordinates:

$$d^3p_3 = |\bar{p_3}|^2 d|\bar{p_3}|\sin\theta d\theta d\varphi = E_3^2 dE_3 \sin\theta d\theta d\varphi,$$

and from Law of Cosines:

$$|\bar{p_3}+\bar{p_4}| = \sqrt{E_3^2 + E_4^2 + 2E_3E_4 cos\theta},$$

introducing new variable u:

$$\begin{aligned} u = \sqrt{E_3^2 + E_4^2 + 2E_3E_4 cos\theta} = |\bar{p_3}+\bar{p_4}| \\ du = -\frac{E_3E_4\sin\theta d\theta}{\sqrt{E_3^2+E_4^2+2E_3E_4\cos\theta}}, \end{aligned}$$

applying above derivations and integrating over $0 < \varphi < 2\pi$ one obtains:

$$d\Gamma_1 = -2\pi K_1' E_4(m_\mu - E_4)\frac{du dE_3 d^3p_4}{E_4^2}\delta(m_\mu - u - E_3 - E_4),$$

integral over δ reads:

$$-\int_{u_-}^{u_+} du\delta(m_\mu - u - E_3 - E_4) = \int_{u_+}^{u_-} du\delta(m_\mu - u - E_3 - E_4) = \begin{cases} 1 \rightarrow & \text{when } u_- < m_\mu - E_3 - E_4 < u_+ \\ 0 \rightarrow & \text{otherwise} \end{cases},$$

where:

$$\begin{aligned} &\text{for } \theta = 0 \rightarrow u_+ = \sqrt{E_3^2 + E_4^2 + 2E_3E_4} = |E_3 + E_4|, \\ &\text{for } \theta = \pi \rightarrow u_- = \sqrt{E_3^2 + E_4^2 - 2E_3E_4} = |E_3 - E_4|, \end{aligned}$$

then:

$$\pm(E_3 - E_4) < m_\mu - E_3 - E_4 < E_3 + E_4,$$

by taking positive sign one gets:

$$\frac{m_\mu}{2} > E_3 > \frac{m_\mu}{2} - E_4,$$

and by taking negative sign and considering limits for E_3:

$$\frac{m_\mu}{2} > E_4 > 0,$$

one may now integrate over E_3 in above limits, switch d^3p_4 to polar coordinates and then simplify:

$$\begin{aligned} d\Gamma_1 = 2\pi K_1' E_4(m_\mu - 2E_4)\frac{E_4^2 dE_4 \sin\theta d\theta d\varphi}{E_4^2}\int_{m_\mu/2-E_4}^{m_\mu/2} dE_3 = \\ 2\pi K_1'(m_\mu - 2E_4)E_4^2 dE_4 d\varphi \sin\theta d\theta, \end{aligned}$$

in last step, one has to integrate above expression over $0 < \theta < \pi$, $0 < \varphi < 2\pi$ and finally over E_4:

$$d\Gamma_1 = 8\pi^2 K_1' \int_0^{m_\mu/2} (m_\mu - 2E_4)E_4^2 dE_4 = \frac{m_\mu^4 \pi^2 K_1'}{12},$$

one arrives then to the final result:

$$\Gamma_1 = \frac{m_\mu^5 K_1}{2^{10}\pi^3 12}. \tag{3.76}$$

Subsequent part of decay rate has the form:

$$\begin{aligned} d\Gamma_2 = \frac{|\bar{\mathscr{M}}_2|^2}{2m_\mu}\frac{d^3p_2}{(2\pi)^3 2E_2}\frac{d^3p_3}{(2\pi)^3 2E_3}\frac{d^3p_4}{(2\pi)^3 2E_4}(2\pi)^4\delta^4(p_1 - p_2 - p_3 - p_4) = \\ K_2\frac{(p_3 \cdot p_4)(p_3 \cdot p_2)}{2m_\mu}\frac{d^3p_2}{(2\pi)^3 2E_2}\frac{d^3p_3}{(2\pi)^3 2E_3}\frac{d^3p_4}{(2\pi)^3 2E_4}(2\pi)^4\delta^4(p_1 - p_2 - p_3 - p_4), \end{aligned} \tag{3.77}$$

considering previous calculations:

$$\begin{aligned} 2(p_3 \cdot p_2) = p_1^2 - 2(p_1 \cdot p_4) = m_\mu^2 - 2m_\mu E_4, \\ 2(p_3 \cdot p_4) = (p_1 - p_2)^2 = p_1^2 - 2(p_1 \cdot p_2) = m_\mu^2 - 2m_\mu E_2 \end{aligned}$$

then eq. (3.77) becomes:

$$d\Gamma_2 = K_2'(m_\mu - 2E_2)(m_\mu - 2E_4)\frac{d^3p_2}{E_2}\frac{d^3p_3}{E_3}\frac{d^3p_4}{E_4}\delta^4(p_1 - p_2 - p_3 - p_4),$$

with

$$K_2' = \frac{m_\mu K_2}{2^{11}\pi^5}.$$

Integration over d^3p_3 gives:

$$d\Gamma_2 = K_2'(m_\mu - 2E_2)(m_\mu - 2E_4)\frac{d^3p_2 d^3p_4}{|\bar{p_2}+\bar{p_4}|E_2E_4}\delta(m_\mu - E_2 - |\bar{p_2}+\bar{p_4}| - E_4),$$

assuming that $\bar{p_4}$ is fixed, then switching p_2 to polar coordinates, introducing new variable u and integrating over φ leads to the following formula:

$$d\Gamma_2 = -2\pi K_2'(m_\mu - 2E_2)(m_\mu - 2E_4)\frac{dE_2 d^3p_4}{E_4^2}du\delta(m_\mu - E_2 - u - E_4),$$

where:

$$u = \sqrt{E_2^2 + E_4^2 + 2E_2E_4\cos\theta} = |\bar{p_2}+\bar{p_4}|,$$
$$du = -\frac{E_2E_4\sin\theta d\theta}{\sqrt{E_2^2 + E_4^2 + 2E_2E_4 cos\theta}},$$

integral over δ gives boundaries imposed to E_4 and E_2:

$$\frac{m_\mu}{2} > E_2 > \frac{m_\mu}{2} - E_4,$$
$$\frac{m_\mu}{2} > E_4 > 0,$$

one may now integrate over E_2, switch d^3p_4 to polar coordinates and simplify what leads to:

$$d\Gamma_2 = 2\pi K_2'(m_\mu - 2E_4)E_4^2 dE_4 d\varphi \sin\theta d\theta,$$

finally, last integration over $0 < \theta < \pi$, $0 < \varphi < 2\pi$ and E_4 allows to write the final result:

$$\Gamma_2 = \frac{m_\mu^4\pi^2 K_2'}{12} = \frac{m_\mu^5 K_2}{2^{11}\pi^3 12}, \tag{3.78}$$

Last part of decay rate reads:

$$d\Gamma_3 = \frac{|\bar{\mathscr{M}}_3|^2}{2m_\mu}\frac{d^3p_2}{(2\pi)^3 2E_2}\frac{d^3p_3}{(2\pi)^3 2E_3}\frac{d^3p_4}{(2\pi)^3 2E_4}(2\pi)^4\delta^4(p_1 - p_2 - p_3 - p_4) = \tag{3.79}$$
$$K_1\frac{(p_2\cdot p_4)(p_3\cdot p_2)}{2m_\mu}\frac{d^3p_2}{(2\pi)^3 2E_2}\frac{d^3p_3}{(2\pi)^3 2E_3}\frac{d^3p_4}{(2\pi)^3 2E_4}(2\pi)^4\delta^4(p_1 - p_2 - p_3 - p_4),$$

as previously:

$$2(p_3\cdot p_2) = m_\mu^2 - 2m_\mu E_4,$$
$$2(p_2\cdot p_4) = p_1^2 - 2(p_1\cdot p_3) = m_\mu - 2m_\mu E_3$$

then eq. (3.79) reads:

$$d\Gamma_3 = K_3'(m_\mu - 2E_3)(m_\mu - 2E_4)\frac{d^3p_2}{E_2}\frac{d^3p_3}{E_3}\frac{d^3p_4}{E_4}\delta^4(p_1 - p_2 - p_3 - p_4),$$

where:

$$K_3' = \frac{m_\mu K_3}{2^{11}\pi^5}.$$

Integration over d^3p_2 allows to write:

$$d\Gamma_3 = K_3'(m_\mu - 2E_3)(m_\mu - 2E_4)\frac{d^3p_3 d^3p_4}{|\bar{p_3}+\bar{p_4}|E_3E_4}\delta(m_\mu - |\bar{p_3}+\bar{p_4}| - E_3 - E_4),$$

once again, $\bar{p_4}$ is assumed to be fixed, therefore switching p_3 to polar coordinates with new variable u introduced and integration over φ, leads to:

$$d\Gamma_3 = -2\pi K_3'(m_\mu - 2E_3)(m_\mu - 2E_4)\frac{dE_3 d^3p_4}{E_4^2} du\delta(m_\mu - E_3 - u - E_4),$$

with:

$$u = \sqrt{E_3^2 + E_4^2 + 2E_3E_4 cos\theta} = |\bar{p_3} + \bar{p_4}|,$$
$$du = -\frac{E_3E_4 \sin\theta d\theta}{\sqrt{E_3^2 + E_4^2 + 2E_3E_4cos\theta}},$$

limits of E_3 and E_4 are obtained from δ function integral and reads:

$$\frac{m_\mu}{2} > E_3 > \frac{m_\mu}{2} - E_3,$$
$$\frac{m_\mu}{2} > E_4 > 0,$$

now, integration over E_3 and d^3p_4 written in polar coordinates gives:

$$d\Gamma_3 = 2\pi K_3'(m_\mu - 2E_4)E_4^2 dE_4 d\varphi \sin\theta d\theta,$$

in the last step, integration over $0 < \theta < \pi$, $0 < \varphi < 2\pi$ and E_4 brings final result in the following form:

$$\Gamma_3 = \frac{m_\mu^4 \pi^2 K_3'}{12} = \frac{m_\mu^5 K_3}{2^{11}\pi^3 12}. \tag{3.80}$$

Summing up above derivations, the total decay rate of muon decay process in SMEFT from eqs. (3.76, 3.78, 3.80) reads:

$$\Gamma_\mu = \Gamma_1 + \Gamma_2 + \Gamma_3 = \frac{m_\mu^5}{2^{10}\pi^3 12}\left(K_1 + \frac{K_2}{2} + \frac{K_3}{2}\right) = \tag{3.81}$$
$$\frac{m_\mu^5}{2^{10}\pi^3 12}\left(8(D_1^\dagger D_1) + 16(\Re(D_1^\dagger D_{31}) + \Re(D_1^\dagger F_1) + \Re(D_1^\dagger \phi_1)) + \frac{1}{2}8m_\mu \Re(D_1^\dagger \phi_3) - \frac{1}{2}8m_\mu \Re(D_1^\dagger D_{32})\right),$$

with constants defined in Table 3.3.
One can now check agreement of expression (3.81) with Standard Model.
Standard Model part of SMEFT muon decay rate reads:

$$\Gamma_{SM} = 8D_1^\dagger D_1 \frac{m_\mu^5}{2^{10}\pi^3 12} = \frac{m_\mu^5 \bar{g}^4}{(8\pi)^3 12 M_W^4} U_{1I}^\dagger U_{J2} U_{J2}^\dagger U_{1I}, \tag{3.82}$$

in SM, U_{J2} and U_{1I} are unitary, $\bar{g} = g$ and $\bar{G} = G$, assuming this one gets:

$$\Gamma_{SM} = \frac{m_\mu^5 g^2}{\pi^3 192}, \tag{3.83}$$

where:

$$G = \frac{\sqrt{2}g^2}{8M_W^2}, \tag{3.84}$$

above result it identical to one from Standard Model [3].
At this point, one needs to concentrate on constants in result (3.81) that written explicitly reads:

$$8D_1^\dagger D_1 = \frac{2\bar{g}^4}{M_W^4} U_{1I}^\dagger U_{J2} U_{J2}^\dagger U_{1I}, \tag{3.85}$$

$$16\Re(D_1^\dagger D_{31}) = \frac{8\bar{g}^3\sqrt{2}vm_\mu}{M_W^4}\Re(U_{1I}^\dagger U_{J2} U_{Jg_1}^\dagger U_{1I} C_{g_1 2}^{eW}), \tag{3.86}$$

$$16\Re(D_1^\dagger F_1) = \frac{-16\bar{g}^2}{M_W^2}\Re(U_{1I}^\dagger U_{J2} U_{g_4 I} U_{Jg_3}^\dagger C_{12g_3g_4}^{ll}), \tag{3.87}$$

$$16\Re(D_1^\dagger \phi_1) = \frac{2\bar{g}^2}{M_W^2 M_Z^2}\Re(U_{1J}^\dagger U_{J2}(v^2(\bar{g}^2+\bar{g}'^2)(C_{12}^{\varphi l_1}+C_{12}^{\varphi l_3}) - 2\sqrt{2}vm_\mu(\bar{g}C_{12}^{eW}+\bar{g}'C_{12}^{eB}))), \tag{3.88}$$

$$-8m_\mu\Re(D_1^\dagger D_{32}) = -\frac{8m_\mu\bar{g}^3\sqrt{2}v}{M_W^4}\Re(U_{1I}^\dagger U_{J2} U_{Jg_1}^\dagger U_{1I} C_{g_1 2}^{eW}), \tag{3.89}$$

$$8m_\mu\Re(D_1^\dagger \phi_3) = \frac{4m_\mu\sqrt{2}v\bar{g}^2}{M_W^2 M_Z^2}\Re(U_{1J}^\dagger U_{J2}(\bar{g}C_{12}^{eW}+\bar{g}'C_{12}^{eB})). \tag{3.90}$$

From definition (2.15) U_{J2} matrix can be written explicitly as:

$$U_{J2} = [U_{e_L}^\dagger(1+v^2C'^{\varphi l(3)})U_{\nu_L}]_{J2} = [U_{e_L}^\dagger U_{\nu_L} + v^2 U_{e_L}^\dagger C'^{\varphi l(3)})U_{\nu_L}]_{J2},$$

in order to switch $C'^{\varphi l(3)}$ to mass basis one need to use definition (2.16) and since U_{eL} with U_{ν_L} are unitary:

$$U_{J2} = [U_{e_L}^\dagger U_{\nu_L} + v^2 U_{e_L}^\dagger C'^{\varphi l(3)} U_{e_L} U_{e_L}^\dagger U_{\nu_L}]_{J2} =$$
$$[U_{e_L}^\dagger U_{\nu_L} + v^2 C^{\varphi l(3)} U_{e_L}^\dagger U_{\nu_L}] = [(1+v^2C^{\varphi l(3)})U_{e_L}^\dagger U_{\nu_L}]_{J2},$$

following assumption that neutrinos are massless, one can choose such a basis of U_{ν_L} that $U_{e_L}^\dagger U_{\nu_L} = 1$ and then finally:

$$U_{J2} = [1+v^2C^{\varphi l(3)}]_{J2} = \delta_{J2} + v^2 C_{J2}^{\varphi l(3)}. \tag{3.91}$$

All U_{pr} matrices in eqs. (3.85-3.90) can be expressed in the same way as above.
Eq. (3.85) can be therefore rewritten in the following form:

$$8D_1^\dagger D_1 = \frac{2\bar{g}^4}{M_W^4} U_{1I}^\dagger U_{J2} U_{J2}^\dagger U_{1I} = \frac{\bar{g}^4}{4M_W^4}(\delta_{I1}+v^2C_{I1}^{*\varphi l(3)})(\delta_{J2}+v^2C_{J2}^{\varphi l(3)})(\delta_{2J}+v^2C_{2J}^{*\varphi l(3)})(\delta_{1I}+v^2C_{1I}^{\varphi l(3)}),$$

multiplying and rejecting higher order terms one is left with:

$$8D_1^\dagger D_1 = \frac{4\bar{g}^4}{2M_W^4}(\delta_{I1}\delta_{J2}\delta_{2J}\delta_{1I} + \delta_{I1}\delta_{J2}\delta_{2J}v^2C_{1I}^{\varphi l(3)} + \delta_{J2}\delta_{2J}\delta_{1I}v^2C_{I1}^{*\varphi l(3)} + \delta_{2J}\delta_{1I}\delta_{I1}v^2C_{J2}^{\varphi l(3)} + \delta_{I1}\delta_{J2}\delta_{1I}v^2C_{2J}^{*\varphi l(3)}),$$

it is easy to notice that result will be non-zero only if $J=2 \wedge I=1$, therefore applying this remark one arrives to the final result:

$$8D_1^\dagger D_1 = \frac{2\bar{g}^4}{M_W^4}(1+v^2C_{22}^{\varphi l(3)}+v^2C_{22}^{*\varphi l(3)}+v^2C_{11}^{\varphi l(3)}+v^2C_{11}^{*\varphi l(3)}) =$$
$$\frac{2\bar{g}^4}{M_W^4}(1+2v^2\Re(C_{22}^{\varphi l(3)})+2v^2\Re(C_{11}^{\varphi l(3)})). \tag{3.92}$$

Subsequent term, given by eq. (3.78) expressed in a similar way as above reads:

$$16\Re(D_1^\dagger D_{31}) = \frac{8\bar{g}^3\sqrt{2}vm_\mu}{M_W^4}\Re(\delta_{I1}+v^2C_{I1}^{*\varphi l(3)})(\delta_{J2}+v^2C_{J2}^{\varphi l(3)})(\delta_{g_1J}+v^2C_{g_1J}^{*\varphi l(3)})(\delta_{1I}+v^2C_{1I}^{\varphi l(3)}C_{g_12}^{eW}),$$

multiplying and rejecting higher order terms one obtains:

$$16\Re(D_1^\dagger D_{31}) = \frac{8\bar{g}^3\sqrt{2}vm_\mu}{M_W^4}\Re(\delta_{1I}\delta_{J2}\delta_{Jg_1}\delta_{1I}C_{g_12}^{eW}),$$

above expression is non-zero, only if $I=1 \wedge J=2=g_1$ and then the result has form:

$$16\Re(D_1^\dagger D_{31}) = \frac{8\bar{g}^3\sqrt{2}vm_\mu}{M_W^4}\Re(C_{22}^{eW}). \tag{3.93}$$

The same procedure as before in reference to eq. (3.87) gives:

$$16\Re(D_1^\dagger F_1) = \frac{-16\bar{g}^2}{M_W^2}\Re(\delta_{I1}+v^2C_{I1}^{*\varphi l(3)})(\delta_{J2}+v^2C_{J2}^{\varphi l(3)})(\delta_{g_4I}+v^2C_{g_4I}^{*\varphi l(3)})(\delta_{Jg_3}+v^2C_{Jg_3}^{\varphi l(3)}C_{12g_3g_4}^{ll}),$$

then simplifying:

$$16\Re(D_1^\dagger F_1) = \frac{-16\bar{g}^2}{M_W^2}\Re(\delta_{1I}\delta_{J2}\delta_{g_4I}\delta_{Jg_3}C^{ll}_{12g_3g_4}), \tag{3.94}$$

this term is non-zero, only if $I = 1 = g_4 \wedge J = 2 = g_3$ and then one arrives to:

$$16\Re(D_1^\dagger F_1) = \frac{-16\bar{g}^2}{M_W^2}\Re(C^{ll}_{1221}). \tag{3.95}$$

From eq. (3.88) one gets:

$$16\Re(D_1^\dagger \phi_1) = \frac{2\bar{g}^2}{M_W^2 M_Z^2}\Re((\delta_{J1} + v^2 C^{*\varphi l(3)}_{J1})(\delta_{J2} + v^2 C^{\varphi l(3)}_{J2})(v^2(\bar{g}^2 + \bar{g}'^2)(C^{\varphi l_1}_{12} + C^{\varphi l_3}_{12}) - 2\sqrt{2}vm_\mu(\bar{g}C^{eW}_{12} + \bar{g}'C^{eB}_{12})))$$

therefore:

$$16\Re(D_1^\dagger \phi_1) = \frac{2\bar{g}^2}{M_W^2 M_Z^2}\Re(\delta_{J1}\delta_{J2}(v^2(\bar{g}^2 + \bar{g}'^2)(C^{\varphi l_1}_{12} + C^{\varphi l_3}_{12}) - 2\sqrt{2}vm_\mu(\bar{g}C^{eW}_{12} + \bar{g}'C^{eB}_{12}))),$$

since $\delta_{J1}\delta_{J2} = 0$ whole expression vanishes:

$$16\Re(D_1^\dagger \phi_1) = 0. \tag{3.96}$$

Next expression, given by eq. (3.89) reads:

$$-8m_\mu\Re(D_1^\dagger D_{32}) = -\frac{8m_\mu\bar{g}^3\sqrt{2}v}{M_W^4}\Re(\delta_{I1} + v^2 C^{*\varphi l(3)}_{I1})(\delta_{J2} + v^2 C^{\varphi l(3)}_{J2})(\delta_{g_1J} + v^2 C^{*\varphi l(3)}_{g_1J})(\delta_{1I} + v^2 C^{\varphi l(3)}_{1I}C^{eW}_{g_12}),$$

then:

$$-8m_\mu\Re(D_1^\dagger D_{32}) = -\frac{8m_\mu\bar{g}^3\sqrt{2}v}{M_W^4}\Re(\delta_{I1}\delta_{J2}\delta_{g_1J}\delta_{1I}C^{eW}_{g_12}),$$

this expression is only non-zero when $1 = I \wedge J = 2 = g_1$ and then result has following form:

$$-8m_\mu\Re(D_1^\dagger D_{32}) = -\frac{8m_\mu\bar{g}^3\sqrt{2}v}{M_W^4}\Re(C^{eW}_{22}). \tag{3.97}$$

Last term, from eq. (3.90) can be rewritten as:

$$8m_\mu\Re(D_1^\dagger \phi_3) = \frac{4m_\mu\sqrt{2}v\bar{g}^2}{M_W^2 M_Z^2}\Re((\delta_{J1} + v^2 C^{*\varphi l(3)}_{J1})(\delta_{J2} + v^2 C^{\varphi l(3)}_{J2})(\bar{g}C^{eW}_{12} + \bar{g}'C^{eB}_{12}) + \delta_{1J}\delta_{2J}(\bar{g}C^{eW*}_{12} + \bar{g}'C^{eB*}_{12})),$$

then:

$$8m_\mu\Re(D_1^\dagger \phi_3) = \frac{4m_\mu\sqrt{2}v\bar{g}^2}{M_W^2 M_Z^2}\Re(\delta_{1J}\delta_{2J}(\bar{g}C^{eW}_{12} + \bar{g}'C^{eB}_{12})),$$

since $\delta_{J1}\delta_{2J} = 0$ whole expression vanishes:

$$8m_\mu\Re(D_1^\dagger \phi_3) = 0. \tag{3.98}$$

Summing up results given by eqs. (3.5.2-3.98) and noting that all Wilson coefficients are diagonal elements of Hermitian matrices, and are therefore real, constants read:

$$K_1 = 8(D_1^\dagger D_1) + 16(\Re(D_1^\dagger D_{31}) + \Re(D_1^\dagger F_1) + \Re(D_1^\dagger F_1) = \tag{3.99}$$
$$\frac{2\bar{g}^4}{M_W^4}(1 + 2v^2 C^{\varphi l(3)}_{22} + 2v^2 C^{\varphi l(3)}_{11}) + \frac{8\bar{g}^3\sqrt{2}vm_\mu}{M_W^4}C^{eW}_{22} - \frac{16\bar{g}^2}{M_W^2}C^{ll}_{1221},$$

$$K_2 = 0, \tag{3.100}$$

$$K_3 = -8m_\mu\Re(D_1^\dagger D_{32}) = -\frac{8m_\mu\bar{g}^3\sqrt{2}v}{M_W^4}C^{eW}_{22}. \tag{3.101}$$

3.6. Final result for the muon decay width

Final decay rate formula for muon decay in SMEFT reads:

$$\Gamma_\mu = \frac{m_\mu^5}{2^{10}\pi^3 12}(K_1 + \frac{K_3}{2}) = \tag{3.102}$$
$$\frac{m_\mu^5}{2^{10}\pi^3 12}\left(\frac{2\bar{g}^4}{M_W^4}(1+2v^2C_{22}^{\varphi l(3)}+2v^2C_{11}^{\varphi l(3)}) + \frac{8\bar{g}^3\sqrt{2}vm_\mu}{M_W^4}C_{22}^{eW} - \frac{16\bar{g}^2}{M_W^2}C_{1221}^{ll} - \frac{4m_\mu\bar{g}^3\sqrt{2}v}{M_W^4}C_{22}^{eW}\right) =$$
$$\frac{m_\mu^5}{2^{10}\pi^3 12}\left(\frac{2\bar{g}^4}{M_W^4}(1+2v^2C_{22}^{\varphi l(3)}+2v^2C_{11}^{\varphi l(3)}) + \frac{4\bar{g}^3\sqrt{2}vm_\mu}{M_W^4}C_{22}^{eW} - \frac{16\bar{g}^2}{M_W^2}C_{1221}^{ll}\right),$$

using $\bar{G} = \frac{\sqrt{2}\bar{g}^2}{8M_W^2}$ and $\frac{1}{2v^2} = \frac{\bar{g}^2}{8M_W^2}$ the result is simplified to:

$$\Gamma_\mu = \frac{m_\mu^5\bar{G}^2}{192\pi^3}\left(1+2v^2C_{22}^{\varphi l(3)}+2v^2C_{11}^{\varphi l(3)} - 2v^2C_{1221}^{ll} + \frac{2\sqrt{2}vm_\mu}{\bar{g}}C_{22}^{eW}\right). \tag{3.103}$$

Squared Fermi constant G is given then by the following formula:

$$G^2 = \bar{G}^2\left(1+2v^2C_{22}^{\varphi l(3)}+2v^2C_{11}^{\varphi l(3)} - 2v^2C_{1221}^{ll} + \frac{2\sqrt{2}vm_\mu}{\bar{g}}C_{22}^{eW}\right), \tag{3.104}$$

and:

$$G = \bar{G}\sqrt{1+2v^2C_{22}^{\varphi l(3)}+2v^2C_{11}^{\varphi l(3)} - 2v^2C_{1221}^{ll} + \frac{2\sqrt{2}vm_\mu}{\bar{g}}C_{22}^{eW}},$$

expanding the square root one obtains Fermi's constant G in terms of Wilson coefficients within tree level in SMEFT:

$$G = \bar{G}\left(1+v^2C_{22}^{\varphi l(3)}+v^2C_{11}^{\varphi l(3)} - v^2C_{1221}^{ll} + \frac{\sqrt{2}vm_\mu}{\bar{g}}C_{22}^{eW}\right). \tag{3.105}$$

Muon lifetime using eq. (A.36) reads then:

$$\tau_\mu = \frac{1}{\Gamma_{tot}} = \frac{192\pi^3}{m_\mu^5\bar{G}^2}\left(\frac{1}{1+2v^2C_{22}^{\varphi l(3)}+2v^2C_{11}^{\varphi l(3)} - 2v^2C_{1221}^{ll} + \frac{2\sqrt{2}vm_\mu}{\bar{g}}C_{22}^{eW}}\right), \tag{3.106}$$

and after expanding the above expression:

$$\tau_\mu = \frac{192\pi^3}{m_\mu^5\bar{G}^2}\left(1-2v^2C_{22}^{\varphi l(3)}-2v^2C_{11}^{\varphi l(3)} + 2v^2C_{1221}^{ll} - \frac{2\sqrt{2}vm_\mu}{\bar{g}}C_{22}^{eW}\right). \tag{3.107}$$

All Wilson coefficients appearing in results displayed above with comments on their origin are presented in Table 3.6:

Table 3.2: Wilson coefficients present in final result with their origin.

Wilson coefficient	origin
$C_{22}^{\varphi l(3)}$, $C_{11}^{\varphi l(3)}$	corrections to PMNS matrix U_{PMNS} defined by eq. (2.15), present in charged lepton current vertices
C_{22}^{eW}	correction $W^{\pm}$-lepton coupling, present in charged lepton current vertices given by (2.19, 2.20)
C_{1221}^{ll}	corrections to effective 4-lepton contact interaction 2.31, given by (2.23) Feynman rule

Chapter 4

Conclusions

Despite the fact, that the Standard Model describes three of four fundamental interactions (weak interactions, strong interactions, electromagnetism) with great accuracy, its limits and inadequacies forced particle physicists to go beyond SM in search of the more fundamental theory. None of these attempts succeeded (yet at least!) and some interest switched to the Effective Field Theories (EFT's). This approach allows one to parametrize 'New Physics' without deeper insight into fundamental theory. An example of EFT is the Standard Model Effective Field Theory (SMEFT), SM extended with complete set of higher dimension gauge invariant operators, with their coupling strength parametrized by Wilson coefficients C_X. In order to find new phenomena, one has to know where to look for possible deviations from the SM results and it might be achieved by comparing well known processes obtained within SMEFT and SM. Example of such a process is muon decay, particularly important as it is experimentally very accurately measured, and thus very often used as an input quantity in calculational schemes devised to compute other more complicated observables involving SM particles.

In this thesis the muon decay in SMEFT has been recalculated and the following results have been obtained:

1. expression for the total amplitude $\mathscr{M}_{tot}$ of muon decay process in SMEFT, based on diagrams 2.31-2.34, given by formula (3.45) with constants presented in Table 3.3;

2. expression for the square of total amplitude $|\bar{\mathscr{M}}_{tot}|$, averaged over initial and summed over final spins given by formula (3.68);

3. expression for the total decay rate of muon at rest Γ_μ given by formula (3.103) and based on this result: muon lifetime τ_μ eq. (3.107) and corrected Fermi constant G eq. (3.105)

They read as:

$$\Gamma_\mu = \frac{m_\mu^5 \bar{G}^2}{192\pi^3}\left(1 + 2v^2 C_{22}^{\varphi l(3)} + 2v^2 C_{11}^{\varphi l(3)} - 2v^2 C_{1221}^{ll} + \frac{2\sqrt{2} v m_\mu}{\bar{g}} C_{22}^{eW}\right), \tag{4.1}$$

$$\tau_\mu = \frac{192\pi^3}{m_\mu^5 \bar{G}^2}\left(1 - 2v^2 C_{22}^{\varphi l(3)} - 2v^2 C_{11}^{\varphi l(3)} + 2v^2 C_{1221}^{ll} - \frac{2\sqrt{2} v m_\mu}{\bar{g}} C_{22}^{eW}\right), \tag{4.2}$$

$$G = \bar{G}\left(1 + v^2 C_{22}^{\varphi l(3)} + v^2 C_{11}^{\varphi l(3)} - v^2 C_{1221}^{ll} + \frac{\sqrt{2} v m_\mu}{\bar{g}} C_{22}^{eW}\right). \tag{4.3}$$

The obtained expressions for the muon decay has been compared with similar results available in existing literature on the subject, e.g. [6]. The leading terms agree, in addition the first order correction proportional to muon mass, neglected in other publications, has been derived.

Appendix A

Necessary equations and identities

A.1. Gamma matrices (Dirac representation)

$$\gamma^\mu = \{\gamma^0, \gamma^1, \gamma^2, \gamma^3\}, \tag{A.1}$$

$$\gamma_\nu = \eta_{\nu\mu}\gamma^\mu. \tag{A.2}$$

Anticommutation relation:

$$\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}. \tag{A.3}$$

Commutation relation:

$$\sigma^{\mu\nu} = \frac{i}{2}[\gamma^\mu, \gamma^\nu]. \tag{A.4}$$

γ^5 matrix:

$$\gamma^5 = i\gamma^0\gamma^1\gamma^2\gamma^3. \tag{A.5}$$

Properties of γ^5 matrix:

$$(\gamma^5)^\dagger = \gamma^5, \tag{A.6}$$

$$(\gamma^5)^2 = 1, \tag{A.7}$$

$$\{\gamma^5, \gamma^\nu\} = 0. \tag{A.8}$$

It is proper to define projection operators onto left- and right-handed Dirac spinors:

$$P_R = \frac{1+\gamma^5}{2}, \tag{A.9}$$

$$P_L = \frac{1-\gamma^5}{2}. \tag{A.10}$$

Using eq. (A.8) one can prove that:

$$\gamma^\mu P_L = P_R \gamma^\mu. \tag{A.11}$$

Additional properties:

$$P_L P_R = 0, \tag{A.12}$$

$$P_L P_L = P_L, \tag{A.13}$$

$$P_R P_R = P_R. \tag{A.14}$$

Using eqs. (A.1, A.3) one can also prove the following useful identities:

$$\gamma^\mu \gamma_\mu = 4, \tag{A.15}$$

$$\gamma^\mu \gamma^\nu \gamma_\mu = -2\gamma^\nu, \tag{A.16}$$

$$\gamma^\mu \gamma^\nu \gamma^\rho \gamma_\mu = 4\eta^{\nu\rho}, \tag{A.17}$$

$$\gamma^\mu \gamma^\nu \gamma^\rho \gamma^\sigma \gamma_\mu = -2\gamma^\sigma \gamma^\rho \gamma^\nu, \tag{A.18}$$

$$\epsilon^{\alpha\beta\gamma\delta}\epsilon_{\alpha\beta\rho\sigma} = -2(\delta^\mu_\rho \delta^\nu_\sigma - \delta^\mu_\sigma \delta^\nu_\rho). \tag{A.19}$$

A.2. Trace identities

For all gamma matrices γ^μ:

$$\mathrm{tr}(\gamma^\mu) = 0. \tag{A.20}$$

Main properties of trace operator:

$$\mathrm{tr}(A + B) = \mathrm{tr}(A) + \mathrm{tr}(B), \tag{A.21}$$

$$\mathrm{tr}(\alpha A) = \alpha \mathrm{tr}(A), \tag{A.22}$$

$$\mathrm{tr}(ABC) = \mathrm{tr}(CAB) = \mathrm{tr}(BCA). \tag{A.23}$$

Using these properties one can prove the following trace identities:

$$\mathrm{tr}(\text{any odd number of } \gamma^\mu) = 0, \tag{A.24}$$

$$\mathrm{tr}(\gamma^\mu\gamma^\nu) = 4\eta^{\mu\nu}, \tag{A.25}$$

$$\mathrm{tr}(\gamma^\mu\gamma^\nu\gamma^\rho\gamma^\sigma) = 4(\eta^{\mu\nu}\eta^{\rho\sigma} - \eta^{\mu\rho}\eta^{\nu\sigma} + \eta^{\mu\sigma}\eta^{\nu\rho}), \tag{A.26}$$

$$\mathrm{tr}(\gamma^5) = 0, \tag{A.27}$$

$$\mathrm{tr}(\gamma^\mu\gamma^\nu\gamma^5) = 0, \tag{A.28}$$

$$\mathrm{tr}(\gamma^\mu\gamma^\nu\gamma^\rho\gamma^\sigma\gamma^5) = 4i\epsilon^{\mu\nu\rho\sigma}, \tag{A.29}$$

$$\mathrm{tr}(\gamma^\mu\gamma^\nu\gamma^\rho\gamma^\sigma ...) = \mathrm{tr}(...\gamma^\sigma\gamma^\rho\gamma^\nu\gamma^\mu). \tag{A.30}$$

A.3. Dirac equation

Dirac spinors $u(p)$ and $v(p)$ obey Dirac equation:

$$\begin{aligned} (\not{p} - m)u(p) &= 0, \\ \bar{u}(p)(\not{p} - m) &= 0. \end{aligned} \tag{A.31}$$

$$\begin{aligned} \bar{v}(p)(\not{p} + m) &= 0, \\ (\not{p} + m)v(p) &= 0. \end{aligned} \tag{A.32}$$

where $\not{p} = \gamma^\mu p_\mu$ and $\bar{u}(p) = u^\dagger\gamma^0$.
Additionally, polarisation sum formulas are very useful:

$$\sum_s u(p,s)\bar{u}(p,s) = \not{p} + m, \tag{A.33}$$

$$\sum_s v(p,s)\bar{v}(p,s) = \not{p} - m. \tag{A.34}$$

A.4. Decay rate

Decay rate Γ is given by Fermi's golden rule [5], where f and i refers to final and initial states respectively:

$$d\Gamma = \frac{1}{2m_i}\left(\prod_f \frac{d^3p_f}{(2\pi)^3}\frac{1}{2E_f}\right)|\mathscr{M}(m_i \to p_f)|^2(2\pi)^4\delta^{(4)}(p_i - p_f), \tag{A.35}$$

with $\bar{\mathscr{M}}$ as total amplitude, summed over final spins and averaged over initial.
Lifetime of particle is given by the decay rate:

$$\tau = \frac{1}{\Gamma}. \tag{A.36}$$

A.5. Fierz identities

One may also need Fierz matrix identities in order to simplify the result. Below there are presented two of them, which are used during calculations.

$$(P_{L,R})_{ij}(P_{R,L})_{kl} = \frac{1}{2}(\gamma^\mu P_{R,L})_{il}(\gamma_\mu P_{L,R})_{kj}, \tag{A.37}$$

$$(\gamma^\mu P_{L,R})_{ij}(\gamma_\mu P_{L,R})_{kl} = -(\gamma^\mu P_{L,R})_{il}(\gamma_\mu P_{L,R})_{kj}. \tag{A.38}$$

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